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Inventor(s)	Douglas; Jonathan D. et al.

Building data platform with air quality analysis based on mobile air quality sensors

Abstract

A building system operates to receive air quality measurements of a plurality of spaces of the building from a mobile air quality sensor, the mobile air quality sensor configured to collect the air quality measurements of the plurality of spaces while the mobile air quality sensor moves throughout the plurality of spaces of the building and provide the collected air quality measurements to the one or more processors. The air quality measurements include data identifying locations and times at which the air quality measurements were collected The building system operates to cause the one or more processors to store information, or a link to the information, in a database based on the collected air quality measurements, the database including a contextual representation of the plurality of spaces of the building and perform at least one operation using the information based on the air quality measurements.

Inventors: **Douglas; Jonathan D. (Mequon, WI), Drees; Kirk H. (Cedarburg, WI)**

Applicant: **Tyco Fire & Security GmbH** (Neuhausen am Rheinfall, CH)

Family ID: **1000008780965**

Assignee: **TYCO FIRE & SECURITY GMBH (Neuhausen am Rheinfall, CH)**

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Primary Examiner: Collins; Gary

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/237,043 filed Aug. 25, 2021, the entirety of which is incorporated by reference herein.

BACKGROUND
(1) This application relates generally to building systems. This application relates more particularly to sensor systems for buildings. A building can include stationary sensors that measure conditions of the building, for example, temperature. However, a building system that includes stationary sensors may need a stationary sensor in every location of the building to collect data that properly reflects the conditions of a building. Furthermore, the building may not have a storage techniques to relate conditions of the spaces of the building to the actual spaces of the building.
SUMMARY
(2) One implementation of the present disclosure is a building system of a building including one or more memory devices having instructions stored thereon, that, when executed by one or more processors, cause the one or more processors to receive air quality measurements of spaces of the building from a mobile air quality sensor, the mobile air quality sensor configured to collect the air quality measurements of the spaces while the mobile air quality sensor moves throughout the spaces of the building and provide the air quality measurements to the one or more processors. The air quality measurements include data identifying locations and times at which the air quality measurements were collected. The instructions cause the one or more processors to store information, or a link to the information, in a database based on the air quality measurements and perform at least one operation using the information based on the air quality measurements.

(3) In some embodiments, the instructions cause the one or more processors to control operation of a piece of building equipment to control an environmental condition of at least one space of the spaces based on the information of the digital twin.
(4) In some embodiments, the instructions cause the one or more processors to receive locations of the mobile air quality sensor tracking the mobile air quality sensor throughout the spaces of the building, determine, a first set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a first space of the spaces, based on the locations, determine, a second set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a second space of the spaces, based on the locations, store, in the digital twin, first information, or a link to the first information, for the first space based on the first set of air quality measurements, and store, in the digital twin, second information, or a link to the second information, for the second space based on the first set of air quality measurements.
(5) In some embodiments, the mobile air quality sensor includes at least one tractive component configured to move the mobile air quality sensor throughout the spaces of the building.
(6) In some embodiments, the mobile air quality sensor can include a strap configured to be worn by an occupant that moves the mobile air quality sensor throughout the spaces of the building.
(7) In some embodiments, the digital twin includes a building graph including nodes representing entities of the building and edges between the nodes representing relationships between entities of the building. In some embodiments, a first node of the nodes represents the mobile air quality sensor. In some embodiments, a second node of the nodes represents an occupant that carries the mobile air quality sensor throughout the spaces of the building. In some embodiments, an edge of the edges between the first node and the second node indicates that the occupant carries the mobile air quality sensor throughout the spaces of the building.
(8) In some embodiments, the digital twin includes a building graph including nodes representing entities of the building and edges between the nodes representing relationships between entities of the building. In some embodiments, a first node of the nodes represents the mobile air quality sensor. In some embodiments, a second node of the nodes represents an occupant that carries the mobile air quality sensor throughout the spaces of the building. In some embodiments, an edge of the edges between the first node and the second node indicates that the occupant carries the mobile air quality sensor throughout the spaces of the building. In some embodiments, the instructions cause the one or more processors to generate a health metric indicating a health level of the occupant based on the air quality measurements, identify a third node of the nodes related to the second node based on a second edge of the edges between the third node and the second node, and store the health metric, or a link to the health metric, in the third node responsive to identifying the third node.
(9) In some embodiments, the instructions cause the one or more processors to generate a metric describing air quality for a space of the spaces based on the air quality measurements, identify a first node of nodes of a building graph of the digital twin linked to a second node of the nodes representing the space based on an edge of edges between the nodes relating the first node to the second node, and store the metric, or a link to the metric, in the first node responsive to an identification of the first node.
(10) In some embodiments, the instructions cause the one or more processors to receive a timeseries of air quality measurements from the mobile air quality sensor and store the timeseries, or a link to the timeseries, in the digital twin, the digital twin including a representation of the mobile air quality sensor and a relationship between the representation of the mobile air quality sensor and the timeseries or the link to the timeseries.
(11) In some embodiments, the instructions cause the one or more processors to receive a first timeseries of measurements of a first air quality condition from the mobile air quality sensor, receive a second timeseries of measurements of a second air quality condition from the mobile air quality sensor, identify a first node of nodes of a building graph of the digital twin representing the

mobile air quality sensor, identify a second node of the nodes of the building graph of the digital twin representing the first air quality condition based on a first edge of edges of the building graph relating the first node to the second node, store information based on the first timeseries, or a link to the information, in the second node responsive to an identification of the second node, identify a third node of the nodes of the building graph of the digital twin representing the second air quality condition based on a second edge of the edges of the building graph relating the first node to the third node, and store information based on the second timeseries, or a link to the information, in the third node responsive to an identification of the third node.

(12) In some embodiments, the air quality measurements are measurements of at least one of a particulate level, a level of volatile organic compounds, a carbon monoxide level, a carbon dioxide level, and a nitrogen dioxide level.

(13) Another implementation of the present disclosure is a method. The method includes collecting, by a mobile air quality sensor, air quality measurements of spaces while the mobile air quality sensor moves throughout the spaces of a building. The air quality measurements include data identifying locations and times at which the air quality measurements were collected. The method includes providing, by the air quality sensor, the air quality measurements to one or more processing circuits. The method includes receiving, by the one or more processing circuits, the air quality measurements of spaces of the building from the mobile air quality sensor. The method includes storing, by the one or more processing circuits, information, or a link to the information, in a database based on the air quality measurements. The method includes performing, by the one or more processing circuits, at least one operation using the information based on the air quality measurements.

(14) In some embodiments, the method includes controlling, by the one or more processing circuits, operation of a piece of building equipment to control an environmental condition of at least one space of the spaces based on the information of the digital twin.

(15) In some embodiments, the method includes receiving, by the one or more processing circuits, locations of the mobile air quality sensor tracking the mobile air quality sensor throughout the spaces of the building, determining, by the one or more processing circuits, a first set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a first space of the spaces, based on the locations, determining, by the one or more processing circuits, a second set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a second space of the spaces, based on the locations, storing, by the one or more processing circuits, in the digital twin, first information, or a link to the first information, for the first space based on the first set of air quality measurements, and storing, by the one or more processing circuits, in the digital twin, second information, or a link to the second information, for the second space based on the first set of air quality measurements.

(16) In some embodiments, the digital twin including a building graph including nodes representing entities of the building and edges between the nodes representing relationships between entities of the building. In some embodiments, a first node of the nodes represents the mobile air quality sensor. In some embodiments, a second node of the nodes represents an occupant that carries the mobile air quality sensor throughout the spaces of the building. In some embodiments, an edge of the edges between the first node and the second node indicates that the occupant carries the mobile air quality sensor throughout the spaces of the building.

(17) In some embodiments, the digital twin including a building graph including nodes representing entities of the building and edges between the nodes representing relationships between entities of the building. In some embodiments, a first node of the nodes represents the mobile air quality sensor. In some embodiments, a second node of the nodes represents an occupant that carries the mobile air quality sensor throughout the spaces of the building. In some embodiments, an edge of the edges between the first node and the second node indicates that the occupant carries the mobile air quality sensor throughout the spaces of the building. In some

embodiments, the method includes generating, by the one or more processing circuits, a health metric indicating a health level of the occupant based on the air quality measurements, identifying, by the one or more processing circuits, a third node of the nodes related to the second node based on a second edge of the edges between the third node and the second node, and storing, by the one or more processing circuits, the health metric, or a link to the health metric, in the third node responsive to identifying the third node.

(18) In some embodiments, the method includes generating, by the one or more processing circuits, a metric describing air quality for a space of the spaces based on the air quality measurements, identifying, by the one or more processing circuits, a first node of nodes of a building graph of the digital twin linked to a second node of the nodes representing the space based on an edge of edges between the nodes relating the first node to the second node, and storing, by the one or more processing circuits, the metric, or a link to the metric, in the first node responsive to an identification of the first node.

(19) In some embodiments, the method includes receiving, by the one or more processing circuits, a timeseries of air quality measurements from the mobile air quality sensor and storing, by the one or more processing circuits, the timeseries, or a link to the timeseries, in the digital twin, the digital twin including a representation of the mobile air quality sensor and a relationship between the representation of the mobile air quality sensor and the timeseries or the link to the timeseries.

(20) In some embodiments, the method includes receiving, by the one or more processing circuits, a first timeseries of measurements of a first air quality condition from the mobile air quality sensor, receiving, by the one or more processing circuits, a second timeseries of measurements of a second air quality condition from the mobile air quality sensor, identifying, by the one or more processing circuits, a first node of nodes of a building graph of the digital twin representing the mobile air quality sensor, identifying, by the one or more processing circuits, a second node of the nodes of the building graph of the digital twin representing the first air quality condition based on a first edge of edges of the building graph relating the first node to the second node, and storing, by the one or more processing circuits, information based on the first timeseries, or a link to the information, in the second node responsive to an identification of the second node. In some embodiments, the method includes identifying, by the one or more processing circuits, a third node of the nodes of the building graph of the digital twin representing the second air quality condition based on a second edge of the edges of the building graph relating the first node to the third node and storing, by the one or more processing circuits, information based on the second timeseries, or a link to the information, in the third node responsive to an identification of the third node.

(21) Another implementation of the present disclosure is a building system of a building including a mobile air quality sensor, the mobile air quality sensor configured to collect air quality measurements of spaces while the mobile air quality sensor moves throughout the spaces of the building and provide the air quality measurements to one or more processing circuits. The air quality measurements include data identifying locations and times at which the air quality measurements were collected. In some embodiments, the one or more processing circuits configured to receive the air quality measurements of the spaces of the building from the mobile air quality sensor. In some embodiments, the one or more processing circuits store information, or a link to the information, in a database based on the air quality measurements and perform at least one operation using the information based on the air quality measurements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the

accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. **1** is a block diagram of a building data platform including an edge platform, a cloud platform, and a twin manager, according to an exemplary embodiment.

(3) FIG. **2** is a graph projection of the twin manager of FIG. **1** including application programming interface (API) data, capability data, policy data, and services, according to an exemplary embodiment.

(4) FIG. **3** is another graph projection of the twin manager of FIG. **1** including application programming interface (API) data, capability data, policy data, and services, according to an exemplary embodiment.

(5) FIG. **4** is a graph projection of the twin manager of FIG. **1** including equipment and capability data for the equipment, according to an exemplary embodiment.

(6) FIG. **5** is a diagram of an indoor air quality (IAQ) control system including one or more autonomous devices that include sensors, according to some embodiments.

(7) FIG. **6** is a diagram of one of the autonomous devices of FIG. **5**, according to some embodiments.

(8) FIG. **7** is a block diagram of a control system for the IAQ system of FIG. **5** using inputs from one or more of the autonomous devices, according to some embodiments.

(9) FIG. **8** is a block diagram of a system where a mobile air quality sensor collects air quality data of a building that is analyzed by an air quality analyzer, according to an exemplary embodiment.

(10) FIG. **9** is a graph including nodes and edges that represent the mobile air quality sensor of FIG. **8**, timeseries data collected by the mobile air quality sensor of FIG. **8**, and information derived based on the timeseries data, according to an exemplary embodiment.

(11) FIG. **10** is a flow diagram of a process of collecting air quality measurements from a mobile air quality sensor and storing information based on the air quality measurements in a digital twin, according to some embodiments.

DETAILED DESCRIPTION

(12) Referring generally to the FIGURES, a building data platform with air quality analysis based on mobile air quality sensors is shown, according to various exemplary embodiments. A building or other environment can have a variety of air quality conditions, e.g., volatile organic compound (VOC), carbon dioxide (CO_2), total VOC (TVOC), particular matter (PM), etc. that a sensor can measure. However, to effectively identify the air quality throughout the various spaces of the building, the system may need multiple stationary air quality sensors in each area in the building to properly identify the variations in air quality throughout the spaces of the building. The air quality of the spaces of the building may vary substantially across the various areas of the building. Even in a single space of the building (e.g., a long hallway, large conference room, etc.), one stationary sensor may not provide enough spatial resolution. Spatial resolution can be a level of how specific measurements of the sensors are across the building. For example, measurements of ten stationary sensors on a floor of a building may provide a higher spatial resolution of an environmental condition than measurements of three stationary sensors on the floor of the building. Without a high spatial resolution, a source of poor air quality (e.g., wet paint might create high VOC levels) in a particular space such as hallway of a building might not be sensed by a single stationary sensor in the hallway if the stationary sensor is located far enough away from the source of the poor air quality.

(13) If more stationary sensors are installed in a building to improve the spatial resolution, there may be a negative power consumption result. For example, if a large number of stationary sensors are installed throughout the spaces of the building to measure conditions of the spaces with a high spatial resolution, the stationary sensors can collectively consume a high amount of power. Furthermore, as a result of the increased number of stationary sensors, a high level of building

network bandwidth can be consumed by the stationary sensors. For example, each stationary sensor may establish a unique network connection with a data collection system and stream or provide measured data to the data collector. This can result in slow network speeds for the stationary sensors or other devices that communicate on the network. Furthermore, technicians may need to service and replace each of the stationary sensors at different times in the life of the stationary sensors. This can utilize significant amounts of maintenance and diagnostic resources.

(14) The building data platform may have difficulty in determining or identifying the location of each stationary sensor and the space or area of the building that each stationary sensor measures. The stationary sensors of the building may not include localization systems, e.g., GPS systems, Wi-Fi triangulation systems, etc. Therefore, a maintenance individual may need to manually inspect the physical location of each stationary sensor and manually enter the location of the stationary sensor into a system. Often, this manual location of each stationary sensor may not be performed and the building data platform may not store any data indicating where sensors are located within spaces of the building. Furthermore, the database that stores the stationary sensor data may not have any mechanism to relate the stationary sensors to the spaces. With no relation of stationary sensors or measurements of the stationary sensors to spaces of the building there may not be any mechanism to relate the locations of people who occupy the building to the conditions that the stationary sensors measure, e.g., to determine what air quality occupants of the building have been exposed to.

(15) To solve these and other technical challenges, the building data platform described herein can include at least one mobile air quality sensor. Furthermore, the building data platform described herein can use a digital twin of a building to store the measurements of the mobile air quality sensors and execute applications or software module based on data of the digital twin. The mobile air quality sensor can move throughout a building, e.g., move throughout the spaces of the building. The mobile air quality sensor can be a robot, drone, autonomous vacuum cleaner, or other system, device, or apparatus, that moves throughout the building. Furthermore, the mobile air quality sensor can be a wearable device that occupants of the building wear or carry throughout the building. Instead of requiring a stationary sensor to be located in every area of the building, the mobile air quality sensor can be moved throughout the building and provide a high spatial resolution by collecting air quality measurements while the air quality sensor moves through the building. By using mobile air quality sensors, the building platform can operate on a high spatial resolution while requiring fewer sensors.

(16) The mobile air quality sensor can measure, collect, store, or stream air quality measurements of one or multiple different air quality parameters as the mobile air quality sensor moves throughout the various spaces of the building. The mobile air quality sensor can communicate, via a network, with the building data platform to provide the measurements to the building data platform. Furthermore, the mobile air quality sensor can provide (or cause another device to provide) a location of the mobile air quality sensor to the building data platform. The building data platform can use the location data to determine the location within the building that every sensor measurement (or at least a portion of the sensor measurements) were taken at. A single mobile air quality sensor can measure conditions of multiple different locations of the building while only using a limited amount of network resources. For example, instead of requiring multiple simultaneous network connections of multiple different stationary sensors, the single mobile air quality sensor can measure conditions in the various locations that the different stationary sensors would be located and only use a single network connection. The amount of network bandwidth used by the single network connection of the mobile air quality sensor can be less than the amount of network bandwidth of the multiple stationary sensors. Furthermore, the mobile air quality sensor can use less power than the group of stationary air quality sensors collectively consumes while measuring environmental conditions in all of the same locations as the group of stationary air quality sensors. Furthermore, instead of requiring maintenance on the group of stationary air

quality sensors, the single mobile air quality sensor may be the only sensor requiring periodic maintenance and therefore, less maintenance resources can be consumed.

(17) Based on the received air quality measurements and the location data indicating the locations where the measurements were taken within the building, the building data platform can update a digital twin. The digital twin can be a digital twin of a building, space of a building, person, piece of building equipment, etc. The digital twin can be or include a building graph or graph data structure. The digital twin can be or include a building information model, a virtual reality based digital twin, an augmented reality digital twin, or any other digital twin that provides a contextual description of building, the spaces of the building, people of the building, equipment of the building, events of the building, etc. The digital twin can further include measurements of the mobile air quality sensor, locations that the mobile air quality sensor has been located, occupants that carry or wear the mobile air quality sensors, spaces that the occupants of the building are located within, metrics indicating health of the occupants or spaces based on whether the occupant or space has been exposed to the air quality, etc.

(18) The digital twin can include a building graph that represents entities of the building (e.g., the building, spaces of the building, equipment, people, events). The building graph can include edges between nodes of the building graph that indicate relationships between the various entities of the building. The building graph can indicate the locations of various spaces of the building, measurements of the building, locations that occupants have been, etc. to allow applications, pieces of software, or other applications to generate data such as metrics, aggregations, insights, predictions, or inferences for various spaces or people. By using a building graph, the applications that operate against the graph can perform a single query to identify complex relationships between entities of the building and the data associated with the entities of the building. Furthermore, the applications can execute a single query against the building graph instead of needing to query multiple different data sources, e.g., one database for context of a building, another database for telemetry of the mobile air quality sensor, etc. Furthermore, all of the data generated by the applications can be ingested back into the building graph and related to the entities that the generated data describes. For example, if a metric is generated to represent the health of a person, the building graph can use an edge to link a node representing the person to a node representing the metric.

(19) For example, if an application operates to generate a health metric of a user indicating the effect of air quality data on an occupant, the application can generate a single query (or multiple queries) that requests information describing the spaces that the occupant has been located in and the air quality of the spaces that the mobile air quality sensor has been measured. Because the building graph includes the spatial locations of the mobile air quality sensor and the occupant in addition to the actual environmental conditions of the spaces, a single query (or multiple queries) can retrieve all of the necessary data that the application needs to execute. By reducing the number of queries that need to be processed and data sources that need to be queried, the building data platform can execute applications faster by reducing the number of queries necessary which uses less processing resources and less memory and storage resources.

(20) Referring now to FIG. **1**, a building data platform **100** including an edge platform **102**, a cloud platform **106**, and a twin manager **108** are shown, according to an exemplary embodiment. The edge platform **102**, the cloud platform **106**, and the twin manager **108** can each be separate services deployed on the same or different computing systems. In some embodiments, the cloud platform **106** and the twin manager **108** are implemented in off premises computing systems, e.g., outside a building. The edge platform **102** can be implemented on-premises, e.g., within the building. However, any combination of on-premises and off-premises components of the building data platform **100** can be implemented.

(21) The building data platform **100** includes applications **110**. The applications **110** can be various applications that operate to manage the building subsystems **122**. The applications **110** can be

remote or on-premises applications (or a hybrid of both) that run on various computing systems. The applications **110** can include an alarm application **168** configured to manage alarms for the building subsystems **122**. The applications **110** include an assurance application **170** that implements assurance services for the building subsystems **122**. In some embodiments, the applications **110** include an energy application **172** configured to manage the energy usage of the building subsystems **122**. The applications **110** include a security application **174** configured to manage security systems of the building.

(22) In some embodiments, the applications **110** and/or the cloud platform **106** interacts with a user device **176**. In some embodiments, a component or an entire application of the applications **110** runs on the user device **176**. The user device **176** may be a laptop computer, a desktop computer, a smartphone, a tablet, and/or any other device with an input interface (e.g., touch screen, mouse, keyboard, etc.) and an output interface (e.g., a speaker, a display, etc.).

(23) The applications **110**, the twin manager **108**, the cloud platform **106**, and the edge platform **102** can be implemented on one or more computing systems, e.g., on processors and/or memory devices. For example, the edge platform **102** includes processor(s) **118** and memories **120**, the cloud platform **106** includes processor(s) **124** and memories **126**, the applications **110** include processor(s) **164** and memories **166**, and the twin manager **108** includes processor(s) **148** and memories **150**.

(24) The processors can be a general purpose or specific purpose processors, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable processing components. The processors may be configured to execute computer code and/or instructions stored in the memories or received from other computer readable media (e.g., CDROM, network storage, a remote server, etc.).

(25) The memories can include one or more devices (e.g., memory units, memory devices, storage devices, etc.) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. The memories can include random access memory (RAM), read-only memory (ROM), hard drive storage, temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. The memories can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. The memories can be communicably connected to the processors and can include computer code for executing (e.g., by the processors) one or more processes described herein.

(26) The edge platform **102** can be configured to provide connection to the building subsystems **122**. The edge platform **102** can receive messages from the building subsystems **122** and/or deliver messages to the building subsystems **122**. The edge platform **102** includes one or multiple gateways, e.g., the gateways **112-116**. The gateways **112-116** can act as a gateway between the cloud platform **106** and the building subsystems **122**. The gateways **112-116** can be the gateways described in U.S. Provisional Patent Application No. 62/951,897 filed Dec. 20, 2019, the entirety of which is incorporated by reference herein. In some embodiments, the applications **110** can be deployed on the edge platform **102**. In this regard, lower latency in management of the building subsystems **122** can be realized.

(27) The edge platform **102** can be connected to the cloud platform **106** via a network **104**. The network **104** can communicatively couple the devices and systems of building data platform **100**. In some embodiments, the network **104** is at least one of and/or a combination of a Wi-Fi network, a wired Ethernet network, a ZigBee network, a Bluetooth network, and/or any other wireless network. The network **104** may be a local area network or a wide area network (e.g., the Internet, a building WAN, etc.) and may use a variety of communications protocols (e.g., BACnet, IP, LON, etc.). The network **104** may include routers, modems, servers, cell towers, satellites, and/or network switches. The network **104** may be a combination of wired and wireless networks.

(28) The cloud platform **106** can be configured to facilitate communication and routing of messages between the applications **110**, the twin manager **108**, the edge platform **102**, and/or any other system. The cloud platform **106** can include a platform manager **128**, a messaging manager **140**, a command processor **136**, and an enrichment manager **138**. In some embodiments, the cloud platform **106** can facilitate messaging between the building data platform **100** via the network **104**.
(29) The messaging manager **140** can be configured to operate as a transport service that controls communication with the building subsystems **122** and/or any other system, e.g., managing commands to devices (C2D), commands to connectors (C2C) for external systems, commands from the device to the cloud (D2C), and/or notifications. The messaging manager **140** can receive different types of data from the applications **110**, the twin manager **108**, and/or the edge platform **102**. The messaging manager **140** can receive change on value data **142**, e.g., data that indicates that a value of a point has changed. The messaging manager **140** can receive timeseries data **144**, e.g., a time correlated series of data entries each associated with a particular time stamp. Furthermore, the messaging manager **140** can receive command data **146**. All of the messages handled by the cloud platform **106** can be handled as an event, e.g., the data **142-146** can each be packaged as an event with a data value occurring at a particular time (e.g., a temperature measurement made at a particular time).
(30) The cloud platform **106** includes a command processor **136**. The command processor **136** can be configured to receive commands to perform an action from the applications **110**, the building subsystems **122**, the user device **176**, etc. The command processor **136** can manage the commands, determine whether the commanding system is authorized to perform the particular commands, and communicate the commands to the commanded system, e.g., the building subsystems **122** and/or the applications **110**. The commands could be a command to change an operational setting that control environmental conditions of a building, a command to run analytics, etc.
(31) The cloud platform **106** includes an enrichment manager **138**. The enrichment manager **138** can be configured to enrich the events received by the messaging manager **140**. The enrichment manager **138** can be configured to add contextual information to the events. The enrichment manager **138** can communicate with the twin manager **108** to retrieve the contextual information. In some embodiments, the contextual information is an indication of information related to the event. For example, if the event is a timeseries temperature measurement of a thermostat, contextual information such as the location of the thermostat (e.g., what room), the equipment controlled by the thermostat (e.g., what VAV), etc. can be added to the event. In this regard, when a consuming application, e.g., one of the applications **110** receives the event, the consuming application can operate based on the data of the event, the temperature measurement, and also the contextual information of the event.
(32) The enrichment manager **138** can solve a problem that when a device produces a significant amount of information, the information may contain simple data without context. An example might include the data generated when a user scans a badge at a badge scanner of the building subsystems **122**. This physical event can generate an output event including such information as “DeviceBadgeScannerID,” “BadgeID,” and/or “Date/Time.” However, if a system sends this data to a consuming application, e.g., Consumer A and a Consumer B, each customer may need to call the building data platform knowledge service to query information with queries such as, “What space, build, floor is that badge scanner in?” or “What user is associated with that badge?”
(33) By performing enrichment on the data feed, a system can be able to perform inferences on the data. A result of the enrichment may be transformation of the message “DeviceBadgeScannerId, BadgeId, Date/Time,” to “Region, Building, Floor, Asset, DeviceId, BadgeId, UserName, EmployeeId, Date/Time Scanned.” This can be a significant optimization, as a system can reduce the number of calls by 1/n, where n is the number of consumers of this data feed.
(34) By using this enrichment, a system can also have the ability to filter out undesired events. If there are 100 building in a campus that receive 100,000 events per building each hour, but only 1

building is actually commissioned, only 1/10 of the events are enriched. By looking at what events are enriched and what events are not enriched, a system can do traffic shaping of forwarding of these events to reduce the cost of forwarding events that no consuming application wants or reads.
(35) An example of an event received by the enrichment manager **138** may be:
(36) TABLE-US-00001 { “id”: “someguid”, “eventType”: “Device_Heartbeat”, “eventTime”: “2018-01-27T00:00:00+00:00” “eventValue”: 1, “deviceID”: “someguid” }
(37) An example of an enriched event generated by the enrichment manager **138** may be:
(38) TABLE-US-00002 { “id”: “someguid”, “eventType”: “Device_Heartbeat”, “eventTime”: “2018-01-27T00:00:00+00:00” “eventValue”: 1, “deviceID”: “someguid”, “buildingName”: “Building-48”, “buildingID”: “SomeGuid”, “panelID”: “SomeGuid”, “panelName”: “Building-48-Panel-13”, “cityID”: 371, “cityName”: “Milwaukee”, “stateID”: 48, “stateName”: “Wisconsin (WI)”, “countryID”: 1, “countryName”: “United States” }
(39) By receiving enriched events, an application of the applications **110** can be able to populate and/or filter what events are associated with what areas. Furthermore, user interface generating applications can generate user interfaces that include the contextual information based on the enriched events.
(40) The cloud platform **106** includes a platform manager **128**. The platform manager **128** can be configured to manage the users and/or subscriptions of the cloud platform **106**. For example, what subscribing building, user, and/or tenant utilizes the cloud platform **106**. The platform manager **128** includes a provisioning service **130** configured to provision the cloud platform **106**, the edge platform **102**, and the twin manager **108**. The platform manager **128** includes a subscription service **132** configured to manage a subscription of the building, user, and/or tenant while the entitlement service **134** can track entitlements of the buildings, users, and/or tenants.
(41) The twin manager **108** can be configured to manage and maintain a digital twin. The digital twin can be a digital representation of the physical environment, e.g., a building. The twin manager **108** can include a change feed generator **152**, a schema and ontology **154**, a projection manager **156**, a policy manager **158**, an entity, relationship, and event database **160**, and a graph projection database **162**.
(42) The graph projection manager **156** can be configured to construct graph projections and store the graph projections in the graph projection database **162**. Examples of graph projections are shown in FIGS. **11-13**. Entities, relationships, and events can be stored in the database **160**. The graph projection manager **156** can retrieve entities, relationships, and/or events from the database **160** and construct a graph projection based on the retrieved entities, relationships and/or events. In some embodiments, the database **160** includes an entity-relationship collection for multiple subscriptions.
(43) In some embodiment, the graph projection manager **156** generates a graph projection for a particular user, application, subscription, and/or system. In this regard, the graph projection can be generated based on policies for the particular user, application, and/or system in addition to an ontology specific for that user, application, and/or system. In this regard, an entity could request a graph projection and the graph projection manager **156** can be configured to generate the graph projection for the entity based on policies and an ontology specific to the entity. The policies can indicate what entities, relationships, and/or events the entity has access to. The ontology can indicate what types of relationships between entities the requesting entity expects to see, e.g., floors within a building, devices within a floor, etc. Another requesting entity may have an ontology to see devices within a building and applications for the devices within the graph.
(44) The graph projections generated by the graph projection manager **156** and stored in the graph projection database **162** can be a knowledge graph and is an integration point. For example, the graph projections can represent floor plans and systems associated with each floor. Furthermore, the graph projections can include events, e.g., telemetry data of the building subsystems **122**. The graph projections can show application services as nodes and API calls between the services as

edges in the graph. The graph projections can illustrate the capabilities of spaces, users, and/or devices. The graph projections can include indications of the building subsystems **122**, e.g., thermostats, cameras, VAVs, etc. The graph projection database **162** can store graph projections that keep up a current state of a building.

(45) The graph projections of the graph projection database **162** can be digital twins of a building. Digital twins can be digital replicas of physical entities that enable an in-depth analysis of data of the physical entities and provide the potential to monitor systems to mitigate risks, manage issues, and utilize simulations to test future solutions. Digital twins can play an important role in helping technicians find the root cause of issues and solve problems faster, in supporting safety and security protocols, and in supporting building managers in more efficient use of energy and other facilities resources. Digital twins can be used to enable and unify security systems, employee experience, facilities management, sustainability, etc.

(46) In some embodiments the enrichment manager **138** can use a graph projection of the graph projection database **162** to enrich events. In some embodiments, the enrichment manager **138** can identify nodes and relationships that are associated with, and are pertinent to, the device that generated the event. For example, the enrichment manager **138** could identify a thermostat generating a temperature measurement event within the graph. The enrichment manager **138** can identify relationships between the thermostat and spaces, e.g., a zone that the thermostat is located in. The enrichment manager **138** can add an indication of the zone to the event.

(47) Furthermore, the command processor **136** can be configured to utilize the graph projections to command the building subsystems **122**. The command processor **136** can identify a policy for a commanding entity within the graph projection to determine whether the commanding entity has the ability to make the command. For example, the command processor **136**, before allowing a user to make a command, determine, based on the graph projection database **162**, to determine that the user has a policy to be able to make the command.

(48) In some embodiments, the policies can be conditional based policies. For example, the building data platform **100** can apply one or more conditional rules to determine whether a particular system has the ability to perform an action. In some embodiments, the rules analyze a behavioral based biometric. For example, a behavioral based biometric can indicate normal behavior and/or normal behavior rules for a system. In some embodiments, when the building data platform **100** determines, based on the one or more conditional rules, that an action requested by a system does not match a normal behavior, the building data platform **100** can deny the system the ability to perform the action and/or request approval from a higher level system.

(49) For example, a behavior rule could indicate that a user has access to log into a system with a particular IP address between 8 A.M. through 5 P.M. However, if the user logs in to the system at 7 P.M., the building data platform **100** may contact an administrator to determine whether to give the user permission to log in.

(50) The change feed generator **152** can be configured to generate a feed of events that indicate changes to the digital twin, e.g., to the graph. The change feed generator **152** can track changes to the entities, relationships, and/or events of the graph. For example, the change feed generator **152** can detect an addition, deletion, and/or modification of a node or edge of the graph, e.g., changing the entities, relationships, and/or events within the database **160**. In response to detecting a change to the graph, the change feed generator **152** can generate an event summarizing the change. The event can indicate what nodes and/or edges have changed and how the nodes and edges have changed. The events can be posted to a topic by the change feed generator **152**.

(51) The change feed generator **152** can implement a change feed of a knowledge graph. The building data platform **100** can implement a subscription to changes in the knowledge graph. When the change feed generator **152** posts events in the change feed, subscribing systems or applications can receive the change feed event. By generating a record of all changes that have happened, a system can stage data in different ways, and then replay the data back in whatever order the system

wishes. This can include running the changes sequentially one by one and/or by jumping from one major change to the next. For example, to generate a graph at a particular time, all change feed events up to the particular time can be used to construct the graph.

(52) The change feed can track the changes in each node in the graph and the relationships related to them, in some embodiments. If a user wants to subscribe to these changes and the user has proper access, the user can simply submit a web API call to have sequential notifications of each change that happens in the graph. A user and/or system can replay the changes one by one to reinstitute the graph at any given time slice. Even though the messages are “thin” and only include notification of change and the reference “id/seq id,” the change feed can keep a copy of every state of each node and/or relationship so that a user and/or system can retrieve those past states at any time for each node. Furthermore, a consumer of the change feed could also create dynamic “views” allowing different “snapshots” in time of what the graph looks like from a particular context. While the twin manager **108** may contain the history and the current state of the graph based upon schema evaluation, a consumer can retain a copy of that data, and thereby create dynamic views using the change feed.

(53) The schema and ontology **154** can define the message schema and graph ontology of the twin manager **108**. The message schema can define what format messages received by the messaging manager **140** should have, e.g., what parameters, what formats, etc. The ontology can define graph projections, e.g., the ontology that a user wishes to view. For example, various systems, applications, and/or users can be associated with a graph ontology. Accordingly, when the graph projection manager **156** generates an graph projection for a user, system, or subscription, the graph projection manager **156** can generate a graph projection according to the ontology specific to the user. For example, the ontology can define what types of entities are related in what order in a graph, for example, for the ontology for a subscription of “Customer A,” the graph projection manager **156** can create relationships for a graph projection based on the rule: Region←.fwdarw.Building←.fwdarw.Floor←.fwdarw.Space←.fwdarw.Asset

(54) For the ontology of a subscription of “Customer B,” the graph projection manager **156** can create relationships based on the rule: Building←.fwdarw.Floor←.fwdarw.Asset

(55) The policy manager **158** can be configured to respond to requests from other applications and/or systems for policies. The policy manager **158** can consult a graph projection to determine what permissions different applications, users, and/or devices have. The graph projection can indicate various permissions that different types of entities have and the policy manager **158** can search the graph projection to identify the permissions of a particular entity. The policy manager **158** can facilitate fine grain access control with user permissions. The policy manager **158** can apply permissions across a graph, e.g., if “user can view all data associated with floor **1**” then they see all subsystem data for that floor, e.g., surveillance cameras, HVAC devices, fire detection and response devices, etc.

(56) The twin manager **108** includes a query manager **165** and a twin function manager **167**. The query manger **164** can be configured to handle queries received from a requesting system, e.g., the user device **176**, the applications **110**, and/or any other system. The query manager **165** can receive queries that include query parameters and context. The query manager **165** can query the graph projection database **162** with the query parameters to retrieve a result. The query manager **165** can then cause an event processor, e.g., a twin function, to operate based on the result and the context. In some embodiments, the query manager **165** can select the twin function based on the context and/or perform operates based on the context. In some embodiments, the query manager **165** is configured to perform the operations described with reference to FIGS. **5-10**.

(57) The twin function manager **167** can be configured to manage the execution of twin functions. The twin function manager **167** can receive an indication of a context query that identifies a particular data element and/or pattern in the graph projection database **162**. Responsive to the particular data element and/or pattern occurring in the graph projection database **162** (e.g., based on

a new data event added to the graph projection database **162** and/or change to nodes or edges of the graph projection database **162**, the twin function manager **167** can cause a particular twin function to execute. The twin function can execute based on an event, context, and/or rules. The event can be data that the twin function executes against. The context can be information that provides a contextual description of the data, e.g., what device the event is associated with, what control point should be updated based on the event, etc. The twin function manager **167** can be configured to perform the operations of the FIGS. **11-15**.
(58) Referring now to FIG. **2**, a graph projection **200** of the twin manager **108** including application programming interface (API) data, capability data, policy data, and services is shown, according to an exemplary embodiment. The graph projection **200** includes nodes **202-240** and edges **250-272**. The nodes **202-240** and the edges **250-272** are defined according to the key **201**. The nodes **202-240** represent different types of entities, devices, locations, points, persons, policies, and software services (e.g., API services). The edges **250-272** represent relationships between the nodes **202-240**, e.g., dependent calls, API calls, inferred relationships, and schema relationships (e.g., BRICK relationships).
(59) The graph projection **200** includes a device hub **202** which may represent a software service that facilitates the communication of data and commands between the cloud platform **106** and a device of the building subsystems **122**, e.g., door actuator **214**. The device hub **202** is related to a connector **204**, an external system **206**, and a digital asset "Door Actuator" **208** by edge **250**, edge **252**, and edge **254**.
(60) The cloud platform **106** can be configured to identify the device hub **202**, the connector **204**, the external system **206** related to the door actuator **214** by searching the graph projection **200** and identifying the edges **250-254** and edge **258**. The graph projection **200** includes a digital representation of the "Door Actuator," node **208**. The digital asset "Door Actuator" **208** includes a "DeviceNameSpace" represented by node **207** and related to the digital asset "Door Actuator" **208** by the "Property of Object" edge **256**.
(61) The "Door Actuator" **214** has points and timeseries. The "Door Actuator" **214** is related to "Point A" **216** by a "has_a" edge **260**. The "Door Actuator" **214** is related to "Point B" **218** by a "has_A" edge **258**. Furthermore, timeseries associated with the points A and B are represented by nodes "TS" **220** and "TS" **222**. The timeseries are related to the points A and B by "has_a" edge **264** and "has_a" edge **262**. The timeseries "TS" **220** has particular samples, sample **210** and **212** each related to "TS" **220** with edges **268** and **266** respectively. Each sample includes a time and a value. Each sample may be an event received from the door actuator that the cloud platform **106** ingests into the entity, relationship, and event database **160**, e.g., ingests into the graph projection **200**.
(62) The graph projection **200** includes a building **234** representing a physical building. The building includes a floor represented by floor **232** related to the building **234** by the "has_a" edge from the building **234** to the floor **232**. The floor has a space indicated by the edge "has_a" **270** between the floor **232** and the space **230**. The space has particular capabilities, e.g., is a room that can be booked for a meeting, conference, private study time, etc. Furthermore, the booking can be canceled. The capabilities for the floor **232** are represented by capabilities **228** related to space **230** by edge **280**. The capabilities **228** are related to two different commands, command "book room" **224** and command "cancel booking" **226** related to capabilities **228** by edge **284** and edge **282** respectively.
(63) If the cloud platform **106** receives a command to book the space represented by the node, space **230**, the cloud platform **106** can search the graph projection **200** for the capabilities for the **228** related to the space **230** to determine whether the cloud platform **106** can book the room.
(64) In some embodiments, the cloud platform **106** could receive a request to book a room in a particular building, e.g., the building **234**. The cloud platform **106** could search the graph projection **200** to identify spaces that have the capabilities to be booked, e.g., identify the space **230**

based on the capabilities **228** related to the space **230**. The cloud platform **106** can reply to the request with an indication of the space and allow the requesting entity to book the space **230**.

(65) The graph projection **200** includes a policy **236** for the floor **232**. The policy **236** is related set for the floor **232** based on a “To Floor” edge **274** between the policy **236** and the floor **232**. The policy **236** is related to different roles for the floor **232**, read events **238** via edge **276** and send command **240** via edge **278**. The policy **236** is set for the entity **203** based on has edge **251** between the entity **203** and the policy **236**.

(66) The twin manager **108** can identify policies for particular entities, e.g., users, software applications, systems, devices, etc. based on the policy **236**. For example, if the cloud platform **106** receives a command to book the space **230**. The cloud platform **106** can communicate with the twin manager **108** to verify that the entity requesting to book the space **230** has a policy to book the space. The twin manager **108** can identify the entity requesting to book the space as the entity **203** by searching the graph projection **200**. Furthermore, the twin manager **108** can further identify the edge has **251** between the entity **203** and the policy **236** and the edge **1178** between the policy **236** and the command **240**.

(67) Furthermore, the twin manager **108** can identify that the entity **203** has the ability to command the space **230** based on the edge **1174** between the policy **236** and the edge **270** between the floor **232** and the space **230**. In response to identifying the entity **203** has the ability to book the space **230**, the twin manager **108** can provide an indication to the cloud platform **106**.

(68) Furthermore, if the entity makes a request to read events for the space **230**, e.g., the sample **210** and the sample **212**, the twin manager **108** can identify the edge has **251** between the entity **203** and the policy **236**, the edge **1178** between the policy **236** and the read events **238**, the edge **1174** between the policy **236** and the floor **232**, the “has_a” edge **270** between the floor **232** and the space **230**, the edge **268** between the space **230** and the door actuator **214**, the edge **260** between the door actuator **214** and the point A **216**, the “has_a” edge **264** between the point A **216** and the TS **220**, and the edges **268** and **266** between the TS **220** and the samples **210** and **212** respectively.

(69) Referring now to FIG. **3**, a graph projection **300** of the twin manager **108** including application programming interface (API) data, capability data, policy data, and services is shown, according to an exemplary embodiment. The graph projection **300** includes the nodes and edges described in the graph projection **200** of FIG. **2**. The graph projection **300** includes a connection broker **1254** related to capabilities **228** by edge **398***a*. The connection broker **1254** can be a node representing a software application configured to facilitate a connection with another software application. In some embodiments, the cloud platform **106** can identify the system that implements the capabilities **228** by identifying the edge **398***a* between the capabilities **228** and the connection broker **1254**.

(70) The connection broker **1254** is related to an agent that optimizes a space **356** via edge **398***b*. The agent represented by the node **356** can book and cancel bookings for the space represented by the node **230** based on the edge **398***b* between the connection broker **1254** and the node **356** and the edge **398***a* between the capabilities **228** and the connection broker **1254**.

(71) The connection broker **1254** is related to a cluster **308** by edge **398***c*. Cluster **308** is related to connector B **302** via edge **398***e* and connector A **306** via edge **398***d*. The connector A **306** is related to an external subscription service **304**. A connection broker **310** is related to cluster **308** via an edge **311** representing a rest call that the connection broker represented by node **310** can make to the cluster represented by cluster **308**.

(72) The connection broker **310** is related to a virtual meeting platform **312** by an edge **354**. The node **312** represents an external system that represents a virtual meeting platform. The connection broker represented by node **310** can represent a software component that facilitates a connection between the cloud platform **106** and the virtual meeting platform represented by node **312**. When the cloud platform **106** needs to communicate with the virtual meeting platform represented by the node **312**, the cloud platform **106** can identify the edge **354** between the connection broker **310** and the virtual meeting platform **312** and select the connection broker represented by the node **310** to

facilitate communication with the virtual meeting platform represented by the node **312**.

(73) A capabilities node **318** can be connected to the connection broker **310** via edge **360**. The capabilities **318** can be capabilities of the virtual meeting platform represented by the node **312** and can be related to the node **312** through the edge **360** to the connection broker **310** and the edge **354** between the connection broker **310** and the node **312**. The capabilities **318** can define capabilities of the virtual meeting platform represented by the node **312**. The node **320** is related to capabilities **318** via edge **362**. The capabilities may be an invite bob command represented by node **316** and an email bob command represented by node **314**. The capabilities **318** can be linked to a node **320** representing a user, Bob. The cloud platform **106** can facilitate email commands to send emails to the user Bob via the email service represented by the node **304**. The node **304** is related to the connect a node **306** via edge **398*f*** Furthermore, the cloud platform **106** can facilitate sending an invite for a virtual meeting via the virtual meeting platform represented by the node **312** linked to the node **318** via the edge **358**.

(74) The node **320** for the user Bob can be associated with the policy **236** via the “has” edge **364**. Furthermore, the node **320** can have a “check policy” edge **366** with a portal node **324**. The device API node **328** has a check policy edge **370** to the policy node **236**. The portal node **324** has an edge **368** to the policy node **236**. The portal node **324** has an edge **323** to a node **326** representing a user input manager (UIM). The portal node **324** is related to the UIM node **326** via an edge **323**. The UIM node **326** has an edge **323** to a device API node **328**. The UIM node **326** is related to the door actuator node **214** via edge **372**. The door actuator node **214** has an edge **374** to the device API node **328**. The door actuator **214** has an edge **335** to the connector virtual object **334**. The device hub **332** is related to the connector virtual object via edge **380**. The device API node **328** can be an API for the door actuator **214**. The connector virtual object **334** is related to the device API node **328** via the edge **331**.

(75) The device API node **328** is related to a transport connection broker **330** via an edge **329**. The transport connection broker **330** is related to a device hub **332** via an edge **378**. The device hub represented by node **332** can be a software component that hands the communication of data and commands for the door actuator **214**. The cloud platform **106** can identify where to store data within the graph projection **300** received from the door actuator by identifying the nodes and edges between the points **216** and **218** and the device hub node **332**. Similarly, the cloud platform **106** can identify commands for the door actuator that can be facilitated by the device hub represented by the node **332**, e.g., by identifying edges between the device hub node **332** and an open door node **352** and an lock door node **350**. The door actuator **114** has an edge “has mapped an asset” **280** between the node **214** and a capabilities node **348**. The capabilities node **348** and the nodes **352** and **350** are linked by edges **396** and **394**.

(76) The device hub **332** is linked to a cluster **336** via an edge **384**. The cluster **336** is linked to connector A **340** and connector B **338** by edges **386** and the edge **389**. The connector A **340** and the connector B **338** is linked to an external system **344** via edges **388** and **390**. The external system **344** is linked to a door actuator **342** via an edge **392**.

(77) Referring now to FIG. **4**, a graph projection **400** of the twin manager **108** including equipment and capability data for the equipment is shown, according to an exemplary embodiment. The graph projection **400** includes nodes **402-456** and edges **360-498*f***. The cloud platform **106** can search the graph projection **400** to identify capabilities of different pieces of equipment.

(78) A building node **404** represents a particular building that includes two floors. A floor **1** node **402** is linked to the building node **404** via edge **460** while a floor **2** node **406** is linked to the building node **404** via edge **462**. The floor **2** includes a particular room **2023** represented by edge **464** between floor **2** node **406** and room **2023** node **408**. Various pieces of equipment are included within the room **2023**. A light represented by light node **416**, a bedside lamp node **414**, a bedside lamp node **412**, and a hallway light node **410** are related to room **2023** node **408** via edge **466**, edge **472**, edge **470**, and edge **468**.

(79) The light represented by light node **416** is related to a light connector **426** via edge **484**. The light connector **426** is related to multiple commands for the light represented by the light node **416** via edges **484**, **486**, and **488**. The commands may be a brightness setpoint **424**, an on command **425**, and a hue setpoint **428**. The cloud platform **106** can receive a request to identify commands for the light represented by the light **416** and can identify the nodes **424-428** and provide an indication of the commands represented by the node **424-428** to the requesting entity. The requesting entity can then send commands for the commands represented by the nodes **424-428**.
(80) The bedside lamp node **414** is linked to a bedside lamp connector **481** via an edge **413**. The connector **481** is related to commands for the bedside lamp represented by the bedside lamp node **414** via edges **492**, **496**, and **494**. The command nodes are a brightness setpoint node **432**, an on command node **434**, and a color command **436**. The hallway light **410** is related to a hallway light connector **446** via an edge **498***d*. The hallway light connector **446** is linked to multiple commands for the hallway light node **410** via edges **498***g*, **498***f*, and **498***e*. The commands are represented by an on command node **452**, a hue setpoint node **450**, and a light bulb activity node **448**.
(81) The graph projection **400** includes a name space node **422** related to a server A node **418** and a server B node **420** via edges **474** and **476**. The name space node **422** is related to the bedside lamp connector **481**, the bedside lamp connector **444**, and the hallway light connector **446** via edges **482**, **480**, and **478**. The bedside lamp connector **444** is related to commands, e.g., the color command node **440**, the hue setpoint command **438**, a brightness setpoint command **456**, and an on command **454** via edges **498***c*, **498***b*, **498***a*, and **498**.
(82) Referring particularly to FIG. **5**, a system **500** for a building zone **502** is shown, according to some embodiments. The building zone **502** is a room, a space, an area, etc., of a building, a house, a factory, a manufacturing facility, an office space, etc., according to some embodiments. For example, the system **500** may be implemented in a residential setting, a manufacturing setting, an office setting, a school setting, a laboratory setting, a transit setting, etc., according to some embodiments. The building zone **502** is shown as a single space, but may include any number of rooms, zones, sub-rooms, sections of a room, a floor of a building, an aggregation of multiple zones, etc., according to some embodiments.
(83) In some embodiments, the system **500** includes a base **504**, a controller **506**, and one or more mobile devices, such as autonomous devices **510** (e.g., autonomous robots, mobile air quality sensors, mobile devices, drones, autonomous vacuum cleaners, etc., or any other autonomously movable, translatable, flyable, or repositionable devices that can be configured to move about the building zone **502**), that are designed to move around within the building zone **502**. It should be understood that, while the one or more autonomous devices **510** are described herein as autonomous vacuum cleaning robots, this is for illustrative purposes, and should not be understood as limiting. For example, the one or more autonomous devices **510** may include a combination of autonomous vacuum cleaning robots, drones, individuals with smartphones having one or more sensors, etc., according to some embodiments. Further, in some implementations, non-autonomous devices may be used instead of or in addition to autonomous devices **510**. For example, in some implementations, a mobile device such as a ground or air-based device (e.g., car or other wheeled device, drone, etc.) may be moved around the space manually by a user, either via remote control (e.g., radio frequency, WiFi, Bluetooth, etc.) or by manually moving the device around the space (e.g., walking a wheeled or handheld device around the space). While the disclosure below generally discussed autonomous devices, it should be understood that implementations with non-autonomous devices are contemplated by and fall within the scope of the present disclosure.
(84) In some embodiments, the one or more autonomous devices **510** are configured to move about the building zone **502** (e.g., along a path **508**) to perform various operations at different locations about the building zone **502** (e.g., vacuum cleaning operations, sanitization operations, service operations, etc.). The one or more autonomous devices **510** can include a primary mover (e.g., an electric motor) that is configured to drive the autonomous device **510** to move about the building

zone **502** (e.g., by driving wheels, driving a tread, driving a tractive component or element, etc.), according to some embodiments. In some embodiments, the one or more autonomous devices **510** can include one or more additional primary movers or electric motors for performing the various operations (e.g., an electric motor configured to drive a vacuum cleaning apparatus, etc.). In some embodiments, the one or more autonomous devices **510** also include an energy storage device (e.g., a battery, a multiple of battery cells, etc.). In some embodiments, the one or more autonomous devices **510** are configured to use energy stored in the energy storage device to power the primary movers for mobility and/or performing different operations throughout the building zone **502**.

(85) The base **504** may be or include a zone, a charging station, a controller, a processor, etc., according to some embodiments. In some embodiments, the one or more autonomous devices **510** are configured to return to the base **504** (e.g., for charging). The one or more autonomous devices **510** may capture data (e.g., sensor data, position data, route data, etc.) as the autonomous devices **510** travel along their route, and may provide any of the collected data to the base **504** when the autonomous devices **510** return to the base **504**. In some embodiments, the one or more autonomous devices **510** are configured to communicate with the base **504** when the autonomous devices **510** return to the base. In some embodiments, the autonomous devices **510** are configured to communicate wirelessly with the base **504** and/or the controller **506** as the autonomous devices **510** travel about the building zone **502**. The autonomous devices **510** can be configured to communicate with the base **504** and/or the controller **506** using a wireless communications protocol (e.g., Bluetooth, LoRa, Zigbee, WiFi, cellular communications, radio transmission, etc.). In some embodiments, the autonomous devices **510** are configured to connect with a wireless network of a building of the building zone **502** and transmit information (e.g., position data, sensor data, operational data, etc.) to the base **504** and/or the controller **506** as the autonomous devices **510** travel about the building zone **502**. In this way, information or data from the autonomous devices **510** can be provided to the base **504** and/or the controller **506** in real-time or near real-time. In some embodiments, the autonomous device **510** is configured to log and track information obtained along its route (e.g., position data and corresponding sensor data, operational data, etc.) and provide the logged information to the base **504** when the autonomous devices **510** return to the base **504**. In this way, information from the autonomous devices **510** can be obtained intermittently when the autonomous devices **510** return to the base **504** (e.g., for charging).

(86) In some embodiments, the controller **506** and the base **504** are configured to communicate wirelessly using any of the wireless communications protocols described herein. In this way, any information provided to the base **504** by the autonomous devices **510** can also be provided to the controller **506** (e.g., intermittently and/or in real-time as described in greater detail above).

(87) In some embodiments, the system **500** includes a building system **512**. The building system **512** may be a heating, ventilation, or air conditioning (HVAC) system. In some embodiments, the controller **506** is configured to operate the building system **512** to provide heating, cooling, and/or ventilation to the building zone **502**. In some embodiments, the HVAC system is a variable refrigerant flow (VRF) system, a single stage HVAC system, a multi-stage HVAC system, a variable air volume (VAV) HVAC system, etc., or any other system or combination of systems configured to provide heating, cooling, and/or ventilation to the building zone **502**. The HVAC system can include building equipment (e.g., a chiller, a VRF device, a fan, etc.) configured to operate to affect an environmental or variable condition of the building zone **502**. The building system **512** may also be or include a lighting system, an alert system, a fire suppression system, a reporting system, an optimization system, etc., according to some embodiments.

(88) In some embodiments, the one or more autonomous devices **510** include an environmental condition sensor, an IAQ sensor, etc., or an array of one or more sensors. The autonomous devices **510** can obtain, collect, measure, detect, etc., different values of different environmental conditions, such as IAQ, particulate matter, smoke presence, a fire condition, etc., as the autonomous devices **510** travel about the building zone **502**. In some embodiments, the autonomous devices **510** are

configured to provide sensor data and position to the controller **506** and/or the base **504** in real-time, near real-time or in an intermittent basis (e.g., when the autonomous devices **510** return to the base **504**).

(89) In some embodiments, the one or more autonomous devices **510** include sensors configured to measure a comfort condition of the building zone **502**, an IAQ of the building zone **502**, a biological parameter of the building zone **502**, and/or a safety parameter of the building zone **502**. In some embodiments, for example, the one or more autonomous devices **510** include any of or a combination a temperature sensor, a humidity sensor, and/or a sensor configured to measure mean radiant temperature. In some embodiments, the one or more autonomous devices **510** include any of or a combination of a CO2 sensor, a PM 2.5 sensor, a PM 10 sensor, a sensor configured to measure VOC, a sensor configured to measure ozone, a sensor configured to measure ion concentration, and/or a sensor configured to measure light intensity or ultraviolet (UV) light. In some embodiments, the one or more autonomous devices **510** include a carbon monoxide sensor, a flammable gas sensor (e.g., a sensor configured to measure natural gas), a smoke detector (e.g., a sensor configured to measure soot, smoke, etc.), and/or a sensor configured to measure a presence or concentration of industrial chemicals (e.g., gaseous, liquid, particulate matter, etc.). In some embodiments, the one or more autonomous devices **510** include a sensor that is configured to measure, detect, sense, or otherwise sample air for a specific ribonucleic acid (RNA and/or DNA) (e.g., COVID-19 or any other virus). For example, in such embodiments, the one or more autonomous devices **510** may include a sensor that detects for DNA/RNA associated with a particular disease or pathogen to detect presence of the disease or pathogen. In various embodiments, any combination of the sensors discussed herein could be incorporated into the one or more autonomous devices **510** and used in combination or in isolation to detect conditions in the building zone **502**.

(90) Temperature in the building zone **502** may vary based on a wide variety of parameters, according to some embodiments. Obtaining a temperature mapping of the building zone **502** can facilitate achieving uniform comfort throughout the building zone **502**, according to some embodiments. Mean radiant temperature is a component in human thermal comfort, according to some embodiments. Mean radiant temperature is a measure of how warm or cold surrounding walls of a space are. Obtaining mean radian temperature also facilitates achieving uniform comfort throughout the building zone **502**, according to some embodiments. CO2 is a measure of an amount of ventilation that is provided per occupant of the building zone **502**. High levels of CO2 may indicate insufficient ventilation. The CO2 of building zone **502** may vary spatially due to differences in HVAC airflow distribution, according to some embodiments. The PM 2.5 and PM 10 sensors are sensors that can be configured to measure an amount of particles in the air of the building zone **502**. The PM 2.5 sensor can be configured to measure 2.5 micrometer particles, which may be harmful to occupants if inhaled in quantities greater than a particular level. The PM 10 sensor can be configured to measure an amount of 10 micrometer particles that are in the air of the building zone **502** and may be indicative of dust or dirt in the building zone **502**. VOC can be a measure of volatile organic compounds in the building zone **502**. In some embodiments, VOCs may be found in paints, cleaners, or other chemicals. A concentration of VOCs may vary throughout the building zone **502** due to variation in usage of paints, cleaners, chemicals, etc. Ozone can be a byproduct of high voltage electric discharge and may be detected near electronic devices. In some embodiments, air disinfection devices may generate ions for disinfecting the air. Measuring the ionic concentration in the building zone **502** can facilitate ensuring that uniform ionic concentration is achieved throughout the building zone **502**. Carbon monoxide, industrial chemicals, or flammable gases can be “sniffed” out by the autonomous devices **510** by performing a sweep operation to identify a source of the carbon monoxide, the industrial chemicals, or the flammable gases (e.g., to detect gas leaks), according to some embodiments.

(91) In some embodiments, the controller **506** and/or the base **504** are configured to use both

position data obtained from the autonomous devices **510** and corresponding sensor data (e.g., temperature, humidity, IAQ, smoke detection, gas detection, etc.) to generate a map of variation of the sensor data throughout the building zone **502**. In some embodiments, the controller **506** is configured to identify a location (e.g., position) of a maximum or a minimum of the sensor data (e.g., a location in building zone **502** where temperature is at a maximum, a location in building zone **502** where IAQ is lowest, etc.). Similarly, the controller **506** may generate a map of detected presences of particulate matter, specific DNA/RNA, etc., for various sensor data that is binary (e.g., indicating whether a presence is detected or not) and corresponding location for the detected presence.

(92) In some embodiments, the autonomous devices **510** are configured to generate a positional mapping of the building zone **502** (e.g., while exploring the building zone **502** and determining different boundaries of the building zone **502**). In some embodiments, dimensions (e.g. boundaries, size, floorplan, etc.) of the building zone **502** are known and stored in the controller **506**. In some embodiments, the autonomous devices **510** can perform local control to determine the path **508** along which the autonomous devices **510** can travel through the building zone **502**. In some embodiments, the path **508** is determined by the base **504** and/or the controller **506** (e.g., based on sensor data obtained from previous routes of the autonomous devices **510**, based on real-time sensor data and position data obtained from the autonomous devices **510**, etc.).

(93) Referring now to FIG. **6**, one of the autonomous devices **510** is shown, according to some embodiments. The autonomous device **510** includes a body, housing, or frame **602**, a controller **606**, a wireless transceiver **604**, one or more sensors **620**, an apparatus **622**, a position sensor **614**, and an electric motor **624**, according to some embodiments. In some embodiments, the autonomous device **510** includes one or more tractive components or elements **616** that are configured to engage a ground surface **618** of the building zone **502** to drive, transport, rotate, move, etc., the autonomous device **510** about the building zone **502** or through multiple building zones **502**. In some embodiments, the tractive elements **616** are configured to receive a torque output from the electric motor **624** to drive the autonomous device **510** for transportation. In some embodiments in which the autonomous device is a ground-based vehicle, the tractive elements **616** may be or include wheels, tracks, or other elements for moving across the ground. In some embodiments in which the autonomous device **510** is a drone, the tractive elements **616** may be fans or impeller blades that are driven by the electric motor **624** or a set of electric motors.

(94) The controller **606** is shown to include processing circuitry **612** including a processor **608** and memory **610**, according to some embodiments. The processor **608** may be a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. The processor **608** may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. Processor **608** also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function.

(95) Memory **610** (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. Memory **610** may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein. According to an exemplary embodiment, the memory **610** is communicably connected to the processor **608** via processing circuitry **612** and includes computer code for executing (e.g., by

the processing circuit or the processor) the one or more processes described herein.

(96) In some embodiments, the apparatus **622** is configured to perform different functionality of operations of the autonomous device **510** as the autonomous device **510** travels about the building zone **502**. For example, the apparatus **622** may be a host device apparatus, such as a vacuum cleaning apparatus including a fan or an impeller blade, a canister for collected matter, and an electric motor (e.g., similar to the electric motor **624**), according to some embodiments. As another example, the autonomous device **510** may be a drone, and the apparatus **622** may be a camera or other device that is a part of the drone and performs functions of the drone such as navigation and capturing images/video.

(97) In some embodiments, the autonomous device **510** include one or more arrays of the sensors **620**. For example, the sensors **620** may include a temperature sensor, a mean radiant temperature sensor, a humidity sensor, a CO2 sensor, PM 2.5 sensor, a PM 10 sensor, a sensor configured to measure a presence or concentration of one or more VOCs, an ozone sensor, a sensor configured to measure a concentration of a particular ion, a light intensity sensor, a UV light sensor, a sensor configured to measure a presence or concentration of a particular DNA/RNA in an air sample, a carbon monoxide detector, a natural gas sensor, a flammable gas sensor, a smoke detector, an industrial chemical detector, an IAQ sensor, a dirt sensor, an irradiance sensor, an imaging device, etc. In some embodiments, the autonomous device **510** includes a tubular member that extends from a top of the autonomous device **510** and is configured to obtain an air sample at an elevated distance from the autonomous device **510** (e.g., at a height at which an occupant exhales). If the autonomous device **510** is a vacuum cleaning device or robot, the autonomous device **510** may operate a vacuum cleaning apparatus or a fan of the vacuum cleaning apparatus to stir up dirt on the ground surface **618** so that consistent readings are obtained by the sensors **620**.

(98) In this way, the autonomous device **510** can obtain sensor data related to comfort (e.g., temperature, humidity, etc.), IAQ (e.g., particulate matter, CO2, etc.), biological matter (e.g., DNA/RNA, specific virus DNA/RNA, etc.), and/or safety (e.g., carbon monoxide, flammable gases, smoke, industrial chemicals, etc.). Any of the sensor data obtained by the sensors **620** of the autonomous device **510** can be provided to the controller **606**, according to some embodiments. In some embodiments, the controller **606** is configured to also obtain a corresponding position of the autonomous device **510** (e.g., from the position sensor **614**). In some embodiments, the controller **606** may store the sensor readings and/or position data as timeseries data or other data correlating the sensor readings and/or position data with a time at which the sensor and/or position data was collected. In some embodiments, the controller **606** may store the sensor readings and position data in a form that correlates the sensor readings to positions of the autonomous device **510** at which the sensor readings occurred. The controller **606** provides the sensor data and the position data to the controller **506** and/or the base **504** by operating the wireless transceiver **604**, according to some embodiments. In some embodiments, the controller **606** operates the wireless transceiver **604** to provide the sensor data and the position data to the controller **506** in real-time or near-real time. In some embodiments, the controller **606** collects the sensor data and the position as the autonomous device **510** travels along its route and stores the sensor data and the position data in the memory **610**. The controller **606** may then provide the sensor data and the position data to the controller **506** when the autonomous device **510** returns to the base **504**, according to some embodiments.

(99) It should be understood that while the position of the autonomous device **510** is shown obtained from a position sensor (e.g., the position sensor **614**), the position of the autonomous device **510** (e.g., relative position of the autonomous device **510** in the building zone **502**, relative to the base **504**, etc.) may be obtained using different systems or methods. For example, the position of the autonomous device **510** may be obtained using a triangulation technique, according to some embodiments.

(100) Referring now to FIG. **7**, a portion of the system **500** is shown, according to some embodiments. Specifically, FIG. **7** shows a control system of the system **500**, according to some

embodiments. The control system includes the controller **506**, one or more of the autonomous devices **510** (e.g., a fleet of the autonomous devices **510**), an HVAC system **710**, a fire suppression system **712**, a reporting system **714**, a thermostat **716**, an optimization system **718**, etc., or another system or device **720**, according to some embodiments. In some embodiments, the controller **506** is configured to receive sensor data and position data from the fleet of the autonomous devices **510** (e.g., in real-time, near real-time, intermittently, etc.). The controller **506** may generate control signals and/or reporting data (e.g., the sensor data, the position data, a mapping of sensor data, an identification of a location where a virus-specific DNA/RNA is detected, etc.) to any of the HVAC system **710**, the fire suppression system **712**, the reporting system **714**, the thermostat **716**, the optimization system **718**, or any other system or device **720**. In some embodiments, the controller **506** is configured to adjust an operation of any of the HVAC system **710**, the fire suppression system **712**, the reporting system **714**, the thermostat **716**, the optimization system **718**, and/or another system or device **720** using the sensor data and/or the position data obtained from any of or a combination of the autonomous devices **510**.

(101) In some embodiments, the fleet includes autonomous devices **510** deployed in different spaces or different zones of spaces, such that the fleet together covers different areas of a building. In some embodiments, the autonomous devices **510** of the fleet may be configured to communicate data directly to one another (e.g., via peer-to-peer communication, such as wireless communication, e.g., via a mesh networking configuration) or indirectly via the controller **506**. For example, in some embodiments, one autonomous device **510** may detect presence of a certain condition (presence of an air quality condition, such as a sensed air quality level below a threshold, presence of a contaminant or a virus, etc.) and communicate an indication of the detection to another autonomous device **510** and/or to the controller **506** for use in taking action with respect to other areas or zones (e.g., warning occupants of the presence of the air quality/contaminant issue, restricting access to the space, etc.).

(102) The HVAC system **710** can be any system, device, collection of devices, building equipment, etc., that is configured to operate to affect an environmental or variable condition (e.g., temperature, humidity, etc.) of the building zone **502**, according to some embodiments. The fire suppression system **712** may be an alert system (e.g., a fire alarm system, a fire panel, etc.) and/or a fire sprinkler system configured to extinguish a fire to either alert occupants of the building zone **502** regarding a presence of fire in the building zone **502**, and/or to suppress the fire, according to some embodiments. The reporting system **714** may be a remote system that is configured to aggregate, analyze, and report data (e.g., providing graphical user interfaces, etc.) to a user, an occupant, a building administrator, another system, etc., according to some embodiments. The thermostat **716** may be a processing device (e.g., a wall mounted thermostat including a user interface or a touchscreen) that is configured to obtain environmental condition data of the building zone **502** and operate building equipment or the HVAC system **710** of the building zone **502**, according to some embodiments. The optimization system **718** can be a system identification system that is configured to estimate, predict, or calculate one or more thermal properties of the building zone **502**, a model predictive system, or any other optimization system configured to use data to determine one or more high-level control decisions to minimize energy consumption of a building, maximize efficiency of various systems or sub-systems of the building, etc., according to some embodiments. The other systems or devices **720** can include personal computing devices (e.g., technician smartphones), a fault diagnostics system, a remote database, a remote optimization or analytic system, etc., according to some embodiments.

(103) The controller **506** is shown to include processing circuitry **702** including a processor **704** and memory **706**, according to some embodiments. The processor **704** may be a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof

designed to perform the functions described herein. The processor **704** may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. Processor **608** also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function.

(104) Memory **706** (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. Memory **706** may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein. According to an exemplary embodiment, the memory **706** is communicably connected to the processor **704** via processing circuitry **702** and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

(105) The controller **506** can be configured to generate a mapping for any of the sensor data obtained from the autonomous devices **510** using the position data, according to some embodiments. For example, the controller **506** can generate a mapping of variation or presence of any of temperature, humidity, mean radiant temperature, carbon dioxide, PM 2.5 sensor data, PM 10 sensor data, VOC, ozone, ion concentration, light intensity, UV light intensity, carbon monoxide, flammable gas (e.g., natural gas), smoke or soot, industrial chemicals, specific DNA/RNA (e.g., virus-specific DNA/RNA), etc., with regards to position or location of the building zone **502**. For example, the mapping can show positional or locational variation of the temperature, humidity, etc., throughout the building zone **502**, according to some embodiments.

(106) The controller **506** can use the mapping for any of the sensor data with respect to location of the building zone **502** to generate updates, activation signals, control signals, alert signals, display data, diagnostic data, system information (SI) data, etc., or other data related to sensed variables conditions of the building zone **502** for any of the HVAC system **710**, the fire suppression system **712**, the reporting system **714**, the thermostat **716**, the optimization system **718**, and/or any other system or device **720** (e.g., a personal computing device, a technician system, a building automation system, a building administration system, a security system, etc.). For example, if the sensor data obtained from different positions or locations throughout the building zone **502** indicates poor IAQ, the controller **506** can operate the HVAC system **710** to increase ventilation to the building zone **502**, according to some embodiments. Similarly, if the sensor data obtained from different locations or positions in the building zone **502** indicates that a fire is detected at a particular location in the building zone **502**, the controller **506** can provide an alert to the fire suppression system **712**, according to some embodiments.

(107) Referring generally to FIGS. **5-7**, the autonomous devices **510** can facilitate obtaining information regarding IAQ or particulate matter concentration throughout the building zone **502**, according to some embodiments. The autonomous devices **510** can function as mobile or variable position IAQ sensors that are configured to obtain different IAQ sensor data (e.g., by drawing and sampling air samples) throughout the building zone **502**, according to some embodiments. In some embodiments, the controller **506** is a thermostat that is configured to operate the HVAC system **710** and can use IAQ-related sensor data obtained from the autonomous devices **510** to operate the HVAC system **710** to achieve a desired IAQ (e.g., a desired particulate matter concentration in the building zone **502**, a desired CO_2 concentration in the building zone **502**, etc.). Advantageously, using the autonomous devices **510** to obtain IAQ-related sensor data reduces costs that may be associated with installing and maintaining IAQ sensors throughout the building zone **502**. The autonomous devices **510** can thereby facilitate improved clean air in the building zone **502** by obtaining IAQ sensor data from throughout the building zone **502**, and providing the obtained IAQ

sensor data from different positions or locations in the building zone **502** to a thermostat (e.g., the thermostat **716**, the controller **506**, etc.).
(108) In some implementations, utilizing the autonomous devices **510** may allow for more granular/accurate detection of IAQ in the space/building. IAQ systems relying on a fixed network of sensors, even if multiple sensors are provided within the space/zone, can only sense the conditions of the air at or near the fixed locations. In contrast, the autonomous devices **510** can sense the conditions anywhere within the space that the autonomous devices **510** can move. As a result, the autonomous devices **510** can sample IAQ conditions throughout the space. In some implementations, the system may use a combination of fixed sensors and sensors on the autonomous devices **510** to sample conditions, such that the system can obtain more frequent readings, or even substantially continuous readings, of IAQ conditions in locations where the fixed sensors are located and periodic or less frequent readings of IAQ conditions using the autonomous devices **510**.
(109) Referring still to FIGS. **5-7**, in some embodiments, the autonomous devices **510** are dispatched at scheduled intervals to collect sensor data from throughout the building zone **502** (e.g., from different locations in the building zone **502**). For example, the autonomous devices **510** may be dispatched to travel throughout the building zone **502** several times a day to obtain multiple daily mappings of sensor data or conditions with respect to location. Advantageously, obtaining such sensor data and/or mappings can facilitate improved real-time operation of any of the HVAC system **710**, the fire suppression system **712**, the reporting system **714**, the thermostat **716**, the optimization system **718**, and/or any other system or device **720** of the building zone **502** (e.g., a lighting system).
(110) Referring still to FIGS. **5-7**, any of the sensors **620** or arrays of sensors can be electrically, communicably, or fixedly coupled with the autonomous device **510** (e.g., communicably connected with the controller **606** and fixedly coupled with the body **602**) in a modular unit, according to some embodiments. In some embodiments, different modular units can be plugged into a corresponding port of the autonomous device **510** to fixedly couple the modular unit with the autonomous device **510** and communicably couple different sensors of the modular unit with the controller **606**. For example, different modular units (e.g., a residential modular unit including specific sensors that are relevant for a residential application, a manufacturing-specific modular unit including specific sensors that are relevant for the manufacturing-specific application, etc.) can be removably coupled with the corresponding port of the autonomous devices **510**, according to some embodiments. Advantageously, this facilitates ease of removal and installation of sensors depending on the application. Providing the sensors **620** in different modular units also allows the sensors **620** to be retrofit with the autonomous device **510**, according to some embodiments. In some implementations, the controllers **606** of the autonomous devices **510** may be configured to sense the modular unit(s) connected to the autonomous devices **510** and enable/disable functions of the autonomous devices **510** in response, such as by enabling IAQ monitoring functions in response to an IAQ sensor being connected, infection response functions in response to an infection-related DNA/RNA sensor being connected, etc.
(111) Referring still to FIGS. **5-7**, the autonomous devices **510** can operate in a target-based manner to identify a location that a particular condition originates from, according to some embodiments. For example, if the autonomous devices **510** detect an unusual or excessively high or low sensor value (e.g., light intensity and/or UV light intensity indicating a presence of fire, low air quality reading, etc.), the autonomous devices **510** can be dispatched to identify an originating location in the building zone **502** resulting in the unusual, excessively high or low, or otherwise undesired sensor value, according to some embodiments. The autonomous devices **510** can perform a sweep of the building zone **502** to determine a location or a position in the building zone **502** from which the particular matter, the virus-specific DNA/RNA, etc., or any other undesired condition originates. For example, if one or more of the autonomous devices **510** detect a virus-

specific DNA/RNA or a specific chemical presence, the autonomous devices **510** may perform a sweep (e.g. being dispatched from the base **504**) to identify an originating location of the virus-specific DNA/RNA or the specific chemical presence within the building zone **502**, according to some embodiments. Similarly, these features may be implemented to identify a location of a fire in the building zone **502**, according to some embodiments. In some embodiments, the autonomous devices **510** are configured to provide their sensor data and position data to the controller **506** in real-time while performing the sweep of the building zone **502**.

(112) Referring still to FIGS. **5-7**, the autonomous devices **510** can include a sensor configured to detect a presence or concentration of a virus-specific DNA/RNA (e.g., COVID-19), according to some embodiments. In some embodiments, the autonomous devices **510** include a fan that is configured to operate to draw an air sample across a filter. Different matter, quanta, or virus-specific DNA/RNA may be trapped on the filter, and can be sensed or detected, according to some embodiments. In some embodiments, the sensor that is configured to detect the virus-specific DNA/RNA uses existing infrastructure (e.g., a fan of a vacuum cleaning apparatus) of the autonomous devices **510**. In this way, the apparatus **622** may have a dual-use, both for performing an operation and for performing sensing operations.

(113) Referring still to FIGS. **5-7**, the autonomous devices **510** can include one or more aural alert devices (e.g., speakers) and/or one or more visual alert devices (e.g., an array of light emitting diodes (LEDs)), according to some embodiments. In some embodiments, the autonomous devices **510** are configured to operate to provide a visual alert and/or an aural alert to occupants in the building zone **502** in response to the sensor data. For example, if the sensor data indicates that a virus-specific DNA/RNA is detected in the building zone **502** or that a fire or smoke is detected in the building zone **502**, a visual and/or an aural alert can be provided to occupants of the building zone by activating or operating the one or more aural alert devices and/or the one or more visual alert devices.

(114) Controller **506** may, in various embodiments, be a part of a building management system (BMS) or otherwise integrate with and/or communicate with a BMS that controls various systems of the building. The BMS may control HVAC, security, lighting, and/or other systems of the building using the data collected by the autonomous devices **510**, in various implementations. For example, in some implementations, the controller **506** may be configured to transmit requests or control signals to the BMS and/or devices of the BMS to modify operation of the BMS based on a condition of the space sensed by the autonomous devices **510**. In some implementations, in response to detecting an IAQ condition (e.g., IAQ level below a threshold air quality), the controller **506** may cause the BMS to control HVAC devices, such as air handling units (AHUs) and/or variable air volume (VAV) devices to increase a volume of outside air delivered to the space. In some implementations, the controller **506** may additionally or alternatively respond to such an IAQ condition by activating additional filtering to the air in and/or being delivered to the zone, such as by adding new filters and/or applying different filters to the incoming air at an AHU/VAV and/or activating one or more in-zone filtration devices. In some such implementations, the filtration devices may be activated based on the location of the autonomous devices **510** where the IAQ condition was sensed; for example, the system may include multiple in-zone filtration devices, and the in-zone filtration devices closest to the position of the autonomous devices **510** where the IAQ condition was sensed may be activated. Similar functionality could be provided in response to a sensed presence of a virus/pathogen/contagion.

(115) Similarly, the controller **506** may additionally or alternatively control other systems of the BMS based on the sensed condition. In some implementations, the controller **506** may control security devices (door locks, entry gates, etc.) to restrict access to a space where an air quality or contamination issue is detected. In some implementations, the controller **506** may control fire system devices (e.g., sprinkler systems, alarms, etc.) based on detection of smoke or other indications of fire within the space. Various other systems may be controlled based on the sensed

conditions and are contemplated within the scope of the present disclosure.

(116) Referring now to FIG. **8**, a system **800** where a mobile air quality sensor **804** collects air quality data **830** of a building **802** that is analyzed by an air quality analyzer **814** is shown, according to an exemplary embodiment. The building **802** can be a school, a hospital, a factory, a commercial building, an apartment complex, a sky rise, etc. In some embodiments, the system **800** can be applied to non-building environments, e.g., cities, towns, transit systems, etc. One or multiple mobile air quality sensors **804** may move, transport, or travel around various areas of the building **802** making sensor measurements of the various spaces of the building **802**. The mobile air quality sensor **804** could be a drone (e.g., flying quadcopter), a robot (e.g., with legs or wheels), or any other type of mobile device or system. Examples of a robot with air quality sensors can be found in FIGS. **5-7**. The drone or robot can move through various spaces, floors, zones, rooms, or other areas of the building **802**.

(117) The mobile air quality sensor **804** can be a wearable device. An occupant, user, or individual can carry or wear the mobile air quality sensor **804** while the occupant moves, e.g., moves throughout the various spaces of the building **802**. The user can move the mobile air quality sensor **804** through spaces of a building **802** based on the movements of the user. The wearable device can include a strap or connecting component that allows the occupant to carry the wearable device. The strap can allow the user to wear or carry the wearable device. The strap or connecting component can be a necklace, a watch, a lanyard, a ring, a key-chain, a handle, etc. The wearable device, when a user moves through the building **802**, can collect indoor air quality (IAQ) data **830** of the various routes that the user takes through the building **802**. Multiple different individuals, employees, students, teachers, tenants, etc. may move throughout the building **802** with the wearable device while the wearable device measures the air quality data **830** of the various spaces of the building.

(118) The mobile air quality sensor **804** includes a sensor **806**. The sensor **806** can be an air quality sensor. The sensor **806** can measure and collect air quality data **830** such as particulate or particular levels (e.g., PM1, PM2.5, PM10, etc.). The sensor **806** can measure VOCs or TVOCs or levels of VOCs or TVOCs. The sensor **806** can measure temperature, humidity, carbon monoxide or carbon monoxide level, carbon dioxide or carbon dioxide level, air flow, nitrogen dioxide (NO2) or a NO2 level, etc. In some embodiments, the mobile air quality sensor **804** also includes a sound sensor. The sound sensor can measure audio levels (e.g., signal levels) through the building **802**. For example, in a factory, it may be important for employees to wear hearing protection in loud areas of the factory. If an employee wears the sensor **804**, the system **800** can identify areas in the building **802** that the employee should wear hearing protection. Furthermore, the system **800** could identify which employees of the building **802** may have been exposed to loud sounds and/or to loud sounds for a long duration. The system **800** can identify that employees exposed to high doses of loud sound may need medical attention, provided with high quality hearing protection, should work in a quieter location (e.g., indicated by measurements of the sensor **804**), etc. The mobile air quality sensor **804** can include one or multiple sensors **806**. Each sensor **806** can measure a different environmental condition of the building **802**. At least some of the sensors **806** can be duplicative sensors measuring the same environmental condition of the building **802**.

(119) The mobile air quality sensor **804** includes a communication interface **808**. The communication interface **808** can include one or more wired and/or wireless communications interfaces, radios, transceivers, transmitters, receivers, etc. The communication interface **808** can communicate with wireless beacons **810** of a building and/or a user device **812**. The communication interface **808** can communicate the location data **832** and/or the air quality data **830** to the edge platform **102** via the network **104** or communicate the data to the wireless beacons **810** and/or the user device **812**. The user device **812** can be a smartphone that the user who is wearing the sensor **804** carries. The sensor **804** can communicate, e.g., via Bluetooth, Bluetooth Low Energy, Wi-Fi, LoRa, etc., with the user device **812** or the wireless Beacons **810**. The user device **812** can communicate sensor measurements of the sensor **804** and location data **832** collected by

the user device **812** (e.g., via GPS) to the edge platform **102** via one or more networks, e.g., a Wi-Fi network, a 4G network, a 5G network, a 6G network, a LAN, WAN, MAN, or other network. The edge platform **102**, the cloud platform **106**, or the twin manager **108** can receive the location data **832** or the air quality data **830** from the mobile air quality sensor **804**, from the wireless beacons **810**, or the user device **812**.

(120) In some embodiments, the sensor **804** communicates with one or more wireless beacons **810** of the building **802**. The sensor **804** could, in some embodiments, be a key fob, identification card, or other device carried by a user that communicates with the wireless beacons **810**. The wireless beacons **810** can locate or triangulate the sensor **804** (e.g., via Bluetooth signals) based on signals of the wireless beacons **810** sensed by the communication interface **808** to determine the locations of the sensor **804**. The wireless beacons **810** can communicate the measurements for the sensor **804** and/or the determined location of the sensor **804** to the edge platform **102**. Examples of location tracking techniques that can be implemented by the mobile air quality sensor **804** and the wireless beacons **810** can be found in U.S. patent application Ser. No. 17/220,795 filed Apr. 1, 2021.

(121) The air quality data **830** can include measurements collected by the mobile air quality sensor **804**. The mobile air quality sensor **804** can be configured to collect the air quality data **830** by measuring environmental conditions via the sensors **806** while the mobile air quality sensor **804** moves throughout the building **802** or through spaces of the building **802**. The measurements can be stored by the mobile air quality sensor **804** and transmitted or provided to the wireless beacons **810** or the user device **812** for communication to the edge platform **102**. The air quality data **830** can be collected and stored by the mobile air quality sensor **804** until a connection to the edge platform **102**, the wireless beacon **810**, or the user device **812** is available. The air quality data **830** can be event data including measurements and a timestamp. The air quality data **830** can be a time correlated data stream, e.g., a timeseries of values for each environmental condition measured. The air quality data **830** can be any data packet, data message, data file, or other data structure that stores a value indicating a measurement of at least one environmental condition and a time at which the measurement was measured or collected by the mobile air quality sensor **804**.

(122) The edge platform **102** can receive the location data **832** and/or the air quality data **830** from the sensor **804**, e.g., via the wireless beacons **810** and/or the user device **812**. The edge platform **102** can communicate with the cloud platform **106** and/or the twin manager **108** to integrate the received data with the various components of the twin manager **108**, e.g., data enrichment, integration of the data into a building graph **834**. The building graph **834** can be a graph data structure, an entity database, a graph database, or any other semantic data structure. For example, the twin manager **108** can ingest, store, or add information (or a link to the information) into the building graph **834** based on the received air quality data **830** or the location data **832**. The link to the information could be a resource locator referencing another database, e.g., a timeseries database, that stores the values of the information represented by the link. The link, or the information itself, can be stored in nodes of the building graph **834**, e.g., in properties or other data elements of the nodes.

(123) The building graph **834** can include various nodes and edges between the nodes. The nodes can represent entities of the building **802**, e.g., spaces of the building **802**, the building **802** itself, equipment of the building (e.g., the mobile air quality sensor **804** and/or the building subsystems **122**), measured data (e.g., the air quality data **830**, information derived from the air quality data **830** by the air quality analyzer **814**), event data, events, people, or any other entity of the building. The edges between the nodes can represent relationships between the nodes, e.g., spatial relationships or functional relationships. The spatial relationships can indicate how entities are related spatially, e.g., whether a space is located on a floor, whether a piece of equipment is located in a space, etc. The functional relationships can indicate control or operational relationships between the entities, e.g., a controller operates a damper, a data collector collects data from a sensor, an AHU provides air to a VAV, etc. Through the nodes and edges of the building graph **834**,

the building graph **834** can represent contextual representations of the building **802** and the entities of the building **802** and also information based on the air quality data **830**. The air quality analyzer **814** can query the building graph **834** for data, e.g., the nodes or edges representing the contextual information of the building **802** and/or the information stored in the building graph **834** based on the location data **832** or the air quality data **830**. The air quality analyzer **814** can perform operations such as building operations for the building **802** based on the data queried from the building graph **834**, e.g., the contextual data of the building **802**, the location data **832**, and/or the air quality data **830**. The air quality analyzer **814** can operate based on the queried information to perform operations or control environmental conditions of the building by controlling the building subsystems **122**.

(124) The air quality analyzer **814** can receive the location data **832** and/or the air quality data **830** and perform an analysis on the air quality data **830** and location data **832** to generate actions. The actions can be installing equipment, setting policies for the building **802**, operating equipment to implement increased filtration, increased air changes (e.g., bring in more outdoor air in to the building **802** and discharge air from within the building **802** out of the building **802**), etc. The analyzer **814** includes an air quality building map generator **816**, an individual health analyzer **818**, a space health analyzer **820**, a poor air quality source identifier **822**, a credit generator **824**, a building assessment generator **826**, and a building controller **828**. The results of the analysis and/or actions performed by the air quality analyzer **814** can be an improvement to IAQ. Improving IAQ can increase employee productivity within the building **802**. This may offset any additional costs that result from the actions, e.g., control actions, taken by the analyzer **814** to improve the air quality of the building **802**.

(125) In some embodiments, the air quality analyzer **814** can evaluate the air quality associated with movements of employees both within the building **802** and at home. Because employees may work from home and work from the office, the air quality analyzer **814** can track air quality across the various spaces that the employee occupies. In some embodiments, the air quality analyzer **814** can provide the employee with an insurance program that credits the employee for monitoring their IAQ. An insurance portal could further recommend filters, ventilation, etc. for the building **802** and/or personal spaces associate with the employee. In some embodiments in which the sensor **804** is wearable or otherwise moves with the employee, the sensor **804** may provide a very specific individualized indication of the overall air quality experienced by the employee.

(126) The air quality building map generator **816** can be configured to generate a heat map of various spaces of the building **802**. The heat map can indicate, via a color coded heat signature, spaces of the building **802** with higher and lower air quality, e.g., higher or lower levels of VOC, particulate, humidity, etc. The map may be indicative of ventilation through the building **802**, e.g., areas with higher ventilation may have higher air quality while areas with lower ventilation may have lower air quality. The analyzer **814** can determine which areas of the building **802** have the lowest air quality.

(127) The individual health analyzer **818** can be configured to determine health levels for individuals based on the air quality data **830** and/or the location data **832**. For example, the individual health analyzer **818** can be configured to determine a score indicating how much poor air quality an individual has been exposed to within the building **802**. The score can indicate a length of time that a user was exposed to particular levels, VOCs levels, etc. above a particular level. In some embodiments, the score and/or underlying data can be a component of a broader score or assessment that takes employee health and/or air quality into account, such as a building or space health score with multiple component factors. Examples of multiple level health scoring is described in U.S. patent application Ser. No. 17/708,661 filed Mar. 30, 2022.

(128) In some embodiments, the individual health analyzer **818** can determine air quality metrics for certain employee roles and/or employee teams. In some embodiments, a certain employee role may be experiencing higher than normal and/or lower than acceptable levels of air quality. For

example, individuals working in a painting area of a factory may have higher than average VOC levels. The individual health analyzer **818** could identify the role of painting as a dangerous role and/or operate to improve the air quality in areas were individuals of the painter role are working since they may be using paints that give off high levels of VOC. The credit generator **824** can be configured to determine that employees of the painter role may be due a credit to compensate for their exposure to the high levels of VOC.

(129) In some embodiments, the individual health analyzer **818** can identify that certain teams of the building **802** are exposed to poor air quality. Certain teams may be exposed to high levels of particulate, for example. The individual health analyzer **818** can identify the teams exposed to the high levels of particular and generate a recommendation to move the team to a new area of the building **802** so that the team is in a safer environment.

(130) The poor air quality source identifier **822** can be configured to identify sources of poor air quality within the building **802**. For example, certain pieces of equipment may not be operating properly and might be emitting high levels of VOCs. The space health analyzer **820** can correlate the locations of equipment with the air quality map generated by the generator **816** to determine which pieces of equipment may be a source of VOCs.

(131) In some embodiments, the building **802** is a factory environment. Some areas of the factory may have dirty air. The space health analyzer **820** can identify the location data **832** and/or the air quality data **830** to determine areas of the factory that have low air quality and may be dangerous (e.g., due to chemical usage in various areas of the building, due to paints used in the building, etc.). The analyzer **814** can determine to add and/or operate ventilation or filtration equipment in the low air quality areas of the factory to address the air quality issues. Furthermore, the space health analyzer **820** could be configured to determine paths through the building **802** that have high or low air quality.

(132) The credit generator **824** can be configured to determine reimbursement credits for employees. Some employees that are exposed to high levels of chemicals may be owed hazard pay, e.g., reimbursement for the risks that they take. The credit generator **824** can identify a length of time that the employee is exposed to poor air quality can be used by the credit generator **824** to determine the level of reimbursement that the employee is owed.

(133) The building assessment generator **826** can be configured to generate a summary report for the building **802** based on air quality sensors that are temporarily present in the building **802**. For example, the sensors **804** could be temporarily worn or carried by employees of the building **802** (e.g., for a day, week, or month). Furthermore, the sensors **804** could be carried in and/or clipped onto backpacks of students of a school. The generator **826** can generate the report to include information determined by the analyzer **814**. Examples of a temporary air quality analysis with the sensors **804** can be found in U.S. Provisional Application No. 63/230,608 filed Aug. 6, 2021, the entirety of which is incorporated by reference herein.

(134) The building controller **828** can be configured to operate the building subsystems **122** to improve the air quality of the building **802**. The building controller **828** can be configured to operate ventilation equipment, filtration equipment, increase outdoor air, etc. The building controller **828** can determine to operate to improve the air quality of spaces determined by the analyzer **814** to have air quality below particular levels. The building controller **828** can be configured to operate at least one piece of building equipment of the building subsystems **122** based on the location data **832**, the air quality data **830**, the contextual data of the building graph **834**, or any other information. The building controller **828** can control operation of the building subsystems **122** to control a environmental condition of at least one space of the building **802**.

(135) In some embodiments, the building controller **828** can use the location data **832** of the air quality sensors **804** to determine the locations of occupants. The building controller **828** can combine the locations of occupants with the air quality data **830** to determine which spaces require adjustments to their operation and/or need new equipment. In some embodiments, the building

controller **828** can prioritize high occupancy spaces of the building **802** over low occupancy spaces of the building **802**. Furthermore, once occupancy is prioritized, the controller **828** can prioritize low air quality spaces over high air quality spaces.

(136) In some embodiments, the system **800** can be integrated with emergency response systems. For example, standard operating procedure (SOP) pipelines can utilize the location data **832** and/or air quality data **830** of various occupants of the building **802** received from the sensor **804** to respond to an emergency. Examples of SOP pipelines can be found in U.S. Patent Application No. 62/726,811 filed Sep. 4, 2018, U.S. Provisional Application No. 62/830,892 filed Apr. 8, 2019, U.S. patent application Ser. No. 16/559,318 filed Sep. 3, 2019, and U.S. patent application Ser. No. 17/062,003 filed Oct. 2, 2020, the entireties of which are incorporated by reference herein.

(137) In some embodiments, the system **800** can be used to help evacuate individuals from the building **802**. For example, the sensors **804** can transmit location and air quality information to a system and/or devices of individuals evacuating the building (e.g., police, firefighters, etc.). In some embodiments, the system **800** can be used to help evacuate individuals from the building **802** by helping prioritize individuals for evacuation. The air quality data **830** can indicate the risk levels of users, e.g., carbon dioxide levels, carbon monoxide levels, etc. Higher levels of carbon dioxide or carbon monoxide can indicate smoke and fire. Individuals in high levels of smoke can be prioritized for evacuation. In some embodiments, the prioritization can be converted into directions for the individuals performing the evacuation by the SOP pipeline.

(138) Referring now to FIG. **9**, the graph **834** including nodes **902-922** and edges **950-966** that represent the mobile air quality sensor **804**, timeseries data collected by the mobile air quality sensor **804**, and information derived based on the timeseries data is shown, according to an exemplary embodiment. The graph **834** can be a building graph, a digital twin, an entity graph, a graph data structure, a graph database, a database, etc. The graph **834** can be a graph projection of the graph projection database **162** described with reference to FIGS. **1-4**. The graph **834** can be combined with an artificial intelligence or other application (e.g., the air quality analyzer **814** and/or component of the analyzer **814**) to form a digital twin. The digital twin can be a digital twin of a person, of a building, of a space, of the sensor **804**, etc. Graph and twin related concepts can be found in U.S. patent application Ser. No. 17/354,338 filed Jun. 22, 2021, the entirety of which is incorporated by reference herein. The graph **834** can be managed and/or updated by the twin manager **108** as the data of the sensor **804** is received. The graph **834** can store a real-time and/or dynamic contextual representation of a person, building, and/or sensor.

(139) The graph **834** includes an air quality sensor node **906** representing the sensor **804**. The sensor **804** can be worn or carried by a particular employee, user, or occupant represented by the employee node **910**. The employee represented by the employee node **910** can be related to the air quality sensor **804** via the “wearsA” edge **958** between the employee **910** and the sensor node **906**. The edge **958** between the node **906** and the node **910** can indicate that the user wears, caries, or otherwise moves the mobile air quality sensor **804** throughout the spaces of the building **802**. The sensor node **906** can be associated with various timeseries, the timeseries can be a location timeseries **904** indicating the locations that the sensor **804** has been, PM2.5 timeseries **902** indicating PM2.5 measurements made by the air quality sensor **906**, and VOC timeseries **908** indicating VOC measurements made by the air quality sensor **906**. The sensor **906** can measure the events of the timeseries **902**, **904**, and **908** and store them in the graph **834**. The sensor node **906** is related to the timeseries nodes **902**, **904**, and **908** via the “hasA” edges **950**, **952**, and **954**.

(140) The twin manager **108** can receive air quality data **830** for a specific mobile air quality sensor **804**. The twin manager **108** can receive an identifier (e.g., a name, a product number, a serial number, etc.) of the mobile air quality sensor **804**. The twin manager **108** can identify the node **906** representing the mobile air quality sensor **804** that the twin manager **108** has received the air quality data **830** for based on the identifier. For example, the node **906** may store the identifier and the twin manager **108** can identify the node **906** by comparing the received identifier and the

identifier of the node **906** and determining that the identifiers match. The twin manager **108** can identify nodes for storing the air quality data **830** in by identifying nodes **902**, **904**, and **908** related to the node **906** via edges **952**, **954**, and **950**. For example, the air quality data **830** can include a VOC timeseries and a PM2.5 series and the location data **832** can include a location timeseries. Responsive to identifying the edges **950**, **952**, and **954** relating the node **906** representing the mobile air quality sensor **804**, the twin manager can store the VOC timeseries in the node **908** representing the VOC timeseries, store the PPM2.5 timeseries in the node **902** representing the PM2.5 timeseries, and store the location timeseries in the node **904** representing the location timeseries.

(141) The employee node **910** is related to a building node **916** by a “worksA” edge **956** to indicate that the employee works at a specific building. The employee node **910** is related to a total exposure levels node **912** and a health alert node **914** via “hasA” edges **960** and **962**. The analyzer **814** can determine that the employee is associated with the specific sensor **906** via the edge **958** and analyze the timeseries of nodes **902**, **904**, and/or **908** to determine a total exposure levels for the employee. The total exposure levels can be a virtual data point, a data rollup, a timeseries of values, a status indicator, etc. Examples of timeseries analysis is found in U.S. patent application Ser. No. 15/182,579 filed Jun. 14, 2016, the entirety of which is incorporated by reference herein. Furthermore the health alert **914** may indicate that the employee **910** needs medical attention, e.g., has been exposed to such a high level of VOC or particulate (based on the timeseries **902** and/or **908**) that the employee should have a check-up and/or medical review.

(142) For example, the individual health analyzer **818** can query the building graph for the timeseries stored or linked by the nodes **902**, **908**, and/or **904** responsive to identifying an edge **958** between the node **910** representing the employee and the node **906** representing the sensor **804**, and the nodes **902**, **904**, and **908** being related to the node **906** representing the sensor **804** by the edges **952**, **950**, and **954**. This can indicate that the employee represented by the node **910** is exposed to air quality conditions indicated by the timeseries of the nodes **902**, **904**, and **908**. The individual health analyzer **818** can process the timeseries data and determine various health metrics. For example, the health metrics can indicate an overall health of the employee, an alert that the employee has been exposed to poor air quality, a total length of time exposed to poor air quality, or any other metric that can indicate the physical health of the employee. The individual health analyzer **818** can transmit the resulting metric to the twin manager **108**. The twin manager **108** can store the metric into the building graph **834**. For example, the twin manager **108** can identify the node **910** representing the employee that the individual health analyzer **818** generated the metric for. The twin manager **108** can identify a node **912** to store a total exposure level metric in by identifying the edge **960** between the node **910** representing the employee and the node **912** representing the total exposure level. The twin manager **108** can identify a node **914** to store a health alert in by identifying the edge **962** between the node **910** representing the employee and the node **914** representing the health alert.

(143) The building node **916** is related to a space node **922** via an “includes” edge **968**. The space node **922** is related to a safety levels node **918** indicating safety levels for the space via a “hasA” edge **966**. The safety levels **918** can indicate that the space is unsafe, e.g., has a VOC or PM2.5 above a particular level and/or a daily average over a particular level, etc. The VOC source alert **920** can indicate that the space **922** has a particular source of VOCs, e.g., equipment, etc. The analyzer **814** can determine the VOC source alert **920** based on the VOC timeseries **908**. The VOC source alert **920** can be related to the space node **922** via the “hasA” edge **964**.

(144) The space health analyzer **820** can process the air quality data **830** that corresponds to a particular space. The space health analyzer **820** can identify a portion of the air quality data **830** associated with the particular space based on the location data **832**, e.g., the space health analyzer **820** can identify the portion of air quality data **830** recorded by the mobile air quality sensor **804** within the particular space. The space health analyzer **820** can generate metrics describing health

levels associated with the particular space, e.g., health metrics indicating how healthy it is for occupants to be within the space, e.g., whether the space is safe for occupants to be within or dangerous. The space health analyzer **820** can query the building graph **834** for the portion of air quality data **830** corresponding to a particular space represented by the node **922**, receive the portion of the air quality data **830** from the twin manager **108**, and generate the metric based on the portion of the air quality data **830**. The space health analyzer **820** can send the generated health metric for the space back to the twin manager **108** to be stored in the building graph **834**. The twin manager **108** can identify the node **922** representing the space. The twin manager **108** can identify nodes related to the node **922** that store metrics for the space. For example, the twin manager **108** can identify an edge **966** between the node **922** and the node **918**. Responsive to identifying the node **916** being related to the node **922** by the edge **966**, the twin manager **108** can store safety levels (if the metric is a safety level) in the node **918** representing the safety level. Responsive to identifying the node **920** being related to the node **922** by the edge **964**, the twin manager **108** can store a VOC source alert (if the metric is a VOC source alert) in the node **920** representing the VOC source alert.
(145) The twin manager **108** can use the location data **832** to determine what nodes to store the air quality data **830**, or information derived from the air quality data **830** by the air quality analyzer **814**, in. For example, the twin manager **108** can compare the air quality data **830** against the location data **832** to indicate coordinates, rooms, or spaces that the air quality data **830** was measured, recorded, or sensed in. For example, the location data **832** and the air quality data **830** can be linked via timestamps or a single time stamp. The twin manager **108** can sort the air quality data **830** into various sets of air quality data **830**, each set for a specific space. The twin manager **108** can identify the particular space associated with each set of the air quality data **830** and store the set of air quality data **830** or information derived from the air quality data **830** in the building graph **834** related to the particular space. For example, the twin manager **108** can identify the node **922** representing a space that the twin manager **108** has created a set of air quality data **830** for. The twin manager **108** can store the set of air quality data **830**, or a link to the set of air quality data **830**, in a node related to the node **922** by an edge. For example, the twin manager **108** can receive a derived safety level or VOC source alert based on the set of air quality data and store the safety level in the node **918** responsive to identifying the edge **966** between the node **922** and the node **918**. In some embodiments, the twin manager **108** can store the set of air quality data in a node linked to the node **922** representing the space by an edge. The twin manager **108** can store the VOC source alert in the node **920** responsive to identify the edge **964** between the node **922** and the node **920**.
(146) Referring now to FIG. **10**, a process **1000** of collecting air quality measurements from a mobile air quality sensor and storing information based on the air quality measurements in a digital twin is shown, according to some embodiments. The mobile air quality sensor **804** can perform at least one step of the process **1000**. The wireless beacon **810** can perform at least one step of the process **1000**. The user device **812** can perform at least one step of the process **1000**. Furthermore, the twin manager **108**, the cloud platform **106**, and/or the edge platform **102** can perform at least one step of the process **1000**. The air quality analyzer **814**, or a component of the air quality analyzer **814**, can perform at least one step of the process **1000**. The process **1000** includes a step **1002** of collecting air quality measurements while the mobile air quality sensor moves throughout a building. The process **1000** includes a step **1004** of receiving the air quality measurements from the mobile air quality sensor. The process **1000** includes a step **1006** of storing information based on the air quality measurements in a database. The process **1000** includes a step **1008** of retrieving the information from the database and performing one or more building operations with the information.
(147) In step **1002**, the process **1000** can include collecting, by one or more processing circuits, air quality measurements while the mobile air quality sensor moves throughout a building. For

example, one or more processing circuits of the mobile air quality sensors **804** can receive air quality measurements via the sensor **806**. The mobile air quality sensors **804** can continuously or periodically measure at least one air quality condition while the mobile air quality sensor **804** moves throughout the building **802** (e.g., throughout areas, hallways, zones, floors, or spaces of the building **802**). The mobile air quality sensor **804** can store the air quality data **830** resulting from the measurements on one or more memory or storage The mobile air quality sensor **804** can move through the building **802** by operating a tractive component or being carried by a user, occupant, or other person. While the mobile air quality sensor **804** moves through the building **802**, the mobile air quality sensor **804** can generate, record, measure, or store, the location data **832** or the air quality data **830**. The wireless beacons **810** or the user device **812** can receive information from the mobile air quality sensors **804** and generate the location data **832** or the air quality data **830**.

(148) In step **1004**, the process **1000** can include receiving, by one or more processing circuits, air quality measurements from a mobile air quality sensor. For example, the edge platform **102** can receive the air quality data **830** from the wireless beacons **810**, the user device **812**, and/or the mobile air quality sensor **804**. The edge platform **102** can receive the air quality data **830** via the network **104**. Furthermore, the edge platform **102** can receive the location data **832** from the wireless beacons **810**, the user device **812**, and/or the mobile air quality sensor **804**. The edge platform **102** can receive the location data **832** via the network **104**. The edge platform **102** can communicate the location data **832** and/or the air quality data **830** to the cloud platform **106**. The cloud platform **106** can communicate the location data **832** and/or the air quality data **830** to the twin manager **108**. Furthermore, the edge platform **102**, the cloud platform **106**, and/or the twin manager **108** can communicate the location data **832** and/or the air quality data **830** to the air quality analyzer **814**.

(149) In step **1006**, the process **1000** can include storing, by one or more processing circuits, information based on the air quality measurements in a database. For example, the twin manager **108** can store the air quality data **830** within a digital twin. The digital twin can be or include a building graph **834**. The twin manager **108** can store information, or a link to the information, in the building graph **834**. The information can be the air quality data **830** itself, timeseries data of the air quality data **830**, health metrics derived from the air quality data **830**, location data **832** associated with the air quality data **830**, etc. For example, the twin manager **108** can store the air quality data **830** in the digital twin and the air quality analyzer **814** can query the digital twin for the information. The air quality analyzer **814** can generate information, metrics, values, inferences, predictions, alerts, alarms, or any other information based on the retrieved information. The twin manager **108** can receive the generated information from the air quality analyzer **814** and store the generated information.

(150) For example, the twin manager **108** can store the information, or links to the information, in nodes of the building graph **834**. The twin manager **108** can identify nodes to store the information in based on edges of the building graph **834** relating various nodes. For example, for a particular mobile air quality sensor **804**, the twin manager **108** can identify a first node representing the mobile air quality sensor **804** and edges between the first node and other nodes representing storage locations for the measurements of the mobile air quality sensor **804**. The twin manager **108** can store the information in the other nodes responsive to identifying edges between the first node and the other nodes. For example, the twin manager **108** can store a first timeseries of data for a first environmental condition measured by the mobile air quality sensor **804** in one of the other nodes and a second timeseries of data for a second environmental condition measured by the mobile air quality sensor **804** in another one of the other nodes.

(151) In step **1008**, the process **1000** can include retrieving, by one or more processing circuits, information from the database and perform one or more building operations with the information. For example, the air quality analyzer **814** can query the building graph **834** for information by sending a query data structure including query parameters to the twin manager **108**. The twin

manager **108** can execute the query based on the query parameters to identify information in the building graph **834** and return the information to the air quality analyzer **814**. The twin manager **108** can query the building graph **834** to retrieve the location data **832** or the air quality data **830**. The twin manager **108** can provide the location data **832** or the air quality data **830** to the air quality analyzer **814**.

(152) The air quality analyzer **814** can run various analytics or applications against the retrieved location data **832** or the retrieved air quality data **830**. For example, the various components **816-828** can run against the retrieved location data **832** and/or the retrieved air quality data **830**. For example, the building controller **828** can execute against the location data **832** and/or the air quality data **830**. For example, the building controller **828** can execute against the location data **832** and/or the air quality data **830** to control the operation of the building subsystems **122**. For example, the building controller **828** can be configured to perform at least one operation based on the contextual representation of the building graph **834**, the location data **832**, and/or the air quality data **830**. For example, the building controller **828** can generate control settings, control parameters, or other data to cause a piece of building equipment to control the environmental conditions of at least one space of the building **802**. For example, the building controller **828** can be configured to control an environmental condition of the building **802** to improve the air quality levels of the building **802**.

(153) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

(154) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(155) Although the figures show a specific order of method steps, the order of the steps may differ

from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

(156) In various implementations, the steps and operations described herein may be performed on one processor or in a combination of two or more processors. For example, in some implementations, the various operations could be performed in a central server or set of central servers configured to receive data from one or more devices (e.g., edge computing devices/controllers) and perform the operations. In some implementations, the operations may be performed by one or more local controllers or computing devices (e.g., edge devices), such as controllers dedicated to and/or located within a particular building or portion of a building. In some implementations, the operations may be performed by a combination of one or more central or offsite computing devices/servers and one or more local controllers/computing devices. All such implementations are contemplated within the scope of the present disclosure. Further, unless otherwise indicated, when the present disclosure refers to one or more computer-readable storage media and/or one or more controllers, such computer-readable storage media and/or one or more controllers may be implemented as one or more central servers, one or more local controllers or computing devices (e.g., edge devices), any combination thereof, or any other combination of storage media and/or controllers regardless of the location of such devices.

Claims

1. A building system of a building comprising one or more memory devices having instructions stored thereon, that, when executed by one or more processors, cause the one or more processors to: receive air quality measurements of a plurality of spaces of the building from a mobile air quality sensor comprising at least one tractive component to move the mobile air quality sensor throughout the plurality of spaces of the building, the mobile air quality sensor configured to: collect the air quality measurements of the plurality of spaces while the mobile air quality sensor moves throughout the plurality of spaces of the building, the air quality measurements comprising data identifying locations and times at which the air quality measurements were collected; and provide the air quality measurements to the one or more processors; store information, or a link to the information, in a database based on the air quality measurements; detect, based on the information, a condition occurring in a space of the plurality of spaces; and cause, responsive to detecting the condition occurring in the space, the mobile air quality sensor to move, using the at least one tractive component, to the space to perform subsequent air quality measurements.
2. The building system of claim 1, wherein the instructions cause the one or more processors to: control operation of a piece of building equipment to control an environmental condition of at least one space of the plurality of spaces based on the information of the database.
3. The building system of claim 1, wherein the instructions cause the one or more processors to: receive a plurality of locations of the mobile air quality sensor tracking the mobile air quality sensor throughout the plurality of spaces of the building; determine, a first set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a first space of the plurality of spaces, based on the plurality of locations; determine, a second set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a second space of the plurality of spaces, based on the plurality of locations; store, in the database, first information, or a link to the first information, for the first space based on the first set of air quality measurements; and store, in the database, second information, or a link to the second information, for the second space based on the first set of air quality measurements.

4. The building system of claim 1, comprising a second mobile air quality sensor, the second mobile air quality sensor comprising: a strap configured to be worn by an occupant that moves the second mobile air quality sensor throughout the plurality of spaces of the building.
5. The building system of claim 1, comprising a second mobile air quality sensor configured to be carried by an occupant throughout the plurality of spaces of the building, wherein the database is a digital twin comprising a building graph, the building graph comprising: a plurality of nodes representing entities of the building and a plurality of edges between the plurality of nodes representing relationships between the entities of the building; wherein a first node of the plurality of nodes represents the second mobile air quality sensor; wherein a second node of the plurality of nodes represents the occupant that carries the second mobile air quality sensor throughout the plurality of spaces of the building; wherein an edge of the plurality of edges between the first node and the second node indicates that the occupant carries the second mobile air quality sensor throughout the plurality of spaces of the building.
6. The building system of claim 1, comprising a second mobile air quality sensor configured to be carried by an occupant throughout the plurality of spaces of the building, wherein the database is a digital twin comprising a building graph, the building graph comprising: a plurality of nodes representing entities of the building and a plurality of edges between the plurality of nodes representing relationships between the entities of the building; wherein a first node of the plurality of nodes represents the second mobile air quality sensor; wherein a second node of the plurality of nodes represents the occupant that carries the second mobile air quality sensor throughout the plurality of spaces of the building; wherein an edge of the plurality of edges between the first node and the second node indicates that the occupant carries the second mobile air quality sensor throughout the plurality of spaces of the building; and wherein the instructions cause the one or more processors to: generate a health metric indicating a health level of the occupant based on air quality measurements of the second mobile air quality sensor; identify a third node of the plurality of nodes related to the second node based on a second edge of the plurality of edges between the third node and the second node; and store the health metric, or a link to the health metric, in the third node responsive to identifying the third node.
7. The building system of claim 1, wherein the database is a building graph, wherein the instructions cause the one or more processors to: generate a metric describing air quality for the space of the plurality of spaces based on the air quality measurements; identify a first node of a plurality of nodes of the building graph linked to a second node of the plurality of nodes representing the space based on an edge of a plurality of edges between the plurality of nodes relating the first node to the second node; and store the metric, or a link to the metric, in the first node responsive to an identification of the first node.
8. The building system of claim 1, wherein the instructions cause the one or more processors to: receive a timeseries of the air quality measurements from the mobile air quality sensor; and store the timeseries, or a link to the timeseries, in the database, the database comprising a representation of the mobile air quality sensor and a relationship between the representation of the mobile air quality sensor and the timeseries or the link to the timeseries.
9. The building system of claim 1, wherein the database is a building graph, wherein the instructions cause the one or more processors to: receive a first timeseries of measurements of a first air quality condition from the mobile air quality sensor; receive a second timeseries of measurements of a second air quality condition from the mobile air quality sensor; identify a first node of a plurality of nodes of the building graph representing the mobile air quality sensor; identify a second node of the plurality of nodes of the building graph representing the first air quality condition based on a first edge of a plurality of edges of the building graph relating the first node to the second node; store information based on the first timeseries, or a link to the information, in the second node responsive to an identification of the second node; identify a third node of the plurality of nodes of the building graph representing the second air quality condition

based on a second edge of the plurality of edges of the building graph relating the first node to the third node; and store information based on the second timeseries, or a link to the information, in the third node responsive to an identification of the third node.
10. The building system of claim 1, wherein the air quality measurements are measurements of at least one of: a particulate level; a level of volatile organic compounds; a carbon monoxide level; a carbon dioxide level; and a nitrogen dioxide level.
11. A method, comprising: collecting, by a mobile air quality sensor comprising at least one tractive component to move the mobile air quality sensor throughout a plurality of spaces of a building, air quality measurements of the plurality of spaces of the building while the mobile air quality sensor moves throughout the plurality of spaces of the building, the air quality measurements comprising data identifying locations and times at which the air quality measurements were collected; providing, by the mobile air quality sensor, the air quality measurements to one or more processing circuits; receiving, by the one or more processing circuits, the air quality measurements of the plurality of spaces of the building from the mobile air quality sensor; storing, by the one or more processing circuits, information, or a link to the information, in a database based on the air quality measurements; detecting, by the one or more processing circuits, based on the information, a condition occurring in a space of the plurality of spaces; and causing, by the one or more processing circuits, responsive to detecting the condition occurring in the space, the mobile air quality sensor to move, using the at least one tractive component, to the space to perform subsequent air quality measurements.
12. The method of claim 11, comprising: controlling, by the one or more processing circuits, operation of a piece of building equipment to control an environmental condition of at least one space of the plurality of spaces based on the information of the database.
13. The method of claim 11, comprising: receiving, by the one or more processing circuits, a plurality of locations of the mobile air quality sensor tracking the mobile air quality sensor throughout the plurality of spaces of the building; determining, by the one or more processing circuits, a first set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a first space of the plurality of spaces, based on the plurality of locations; determining, by the one or more processing circuits, a second set of air quality measurements of the air quality measurements taken by the mobile air quality sensor in a second space of the plurality of spaces, based on the plurality of locations; storing, by the one or more processing circuits, in the database, first information, or a link to the first information, for the first space based on the first set of air quality measurements; and storing, by the one or more processing circuits, in the database, second information, or a link to the second information, for the second space based on the first set of air quality measurements.
14. The method of claim 11, comprising a second mobile air quality sensor configured to be carried by an occupant throughout the plurality of spaces of the building, wherein the database is a digital twin comprising a building graph, the building graph comprising: a plurality of nodes representing entities of the building and a plurality of edges between the plurality of nodes representing relationships between the entities of the building; wherein a first node of the plurality of nodes represents the second mobile air quality sensor; wherein a second node of the plurality of nodes represents the occupant that carries the second mobile air quality sensor throughout the plurality of spaces of the building; wherein an edge of the plurality of edges between the first node and the second node indicates that the occupant carries the second mobile air quality sensor throughout the plurality of spaces of the building.
15. The method of claim 11, comprising a second mobile air quality sensor configured to be carried by an occupant throughout the plurality of spaces of the building, wherein the database is a digital twin comprising a building graph, the building graph comprising: a plurality of nodes representing entities of the building and a plurality of edges between the plurality of nodes representing relationships between the entities of the building; wherein a first node of the plurality of nodes

represents the second mobile air quality sensor; wherein a second node of the plurality of nodes represents the occupant that carries the second mobile air quality sensor throughout the plurality of spaces of the building; wherein an edge of the plurality of edges between the first node and the second node indicates that the occupant carries the second mobile air quality sensor throughout the plurality of spaces of the building; and wherein the method includes: generating, by the one or more processing circuits, a health metric indicating a health level of the occupant based on air quality measurements of the second mobile air quality sensor; identifying, by the one or more processing circuits, a third node of the plurality of nodes related to the second node based on a second edge of the plurality of edges between the third node and the second node; and storing, by the one or more processing circuits, the health metric, or a link to the health metric, in the third node responsive to identifying the third node.

16. The method of claim 11, wherein the database is a building graph, the method comprising: generating, by the one or more processing circuits, a metric describing air quality for the space of the plurality of spaces based on the air quality measurements; identifying, by the one or more processing circuits, a first node of a plurality of nodes of the building graph linked to a second node of the plurality of nodes representing the space based on an edge of a plurality of edges between the plurality of nodes relating the first node to the second node; and storing, by the one or more processing circuits, the metric, or a link to the metric, in the first node responsive to an identification of the first node.

17. The method of claim 11, comprising: receiving, by the one or more processing circuits, a timeseries of the air quality measurements from the mobile air quality sensor; and storing, by the one or more processing circuits, the timeseries, or a link to the timeseries, in the database, the database comprising a representation of the mobile air quality sensor and a relationship between the representation of the mobile air quality sensor and the timeseries or the link to the timeseries.

18. The method of claim 11, wherein the database is a building graph, the method comprising: receiving, by the one or more processing circuits, a first timeseries of measurements of a first air quality condition from the mobile air quality sensor; receiving, by the one or more processing circuits, a second timeseries of measurements of a second air quality condition from the mobile air quality sensor; identifying, by the one or more processing circuits, a first node of a plurality of nodes of the building graph representing the mobile air quality sensor; identifying, by the one or more processing circuits, a second node of the plurality of nodes of the building graph of the database representing the first air quality condition based on a first edge of a plurality of edges of the building graph relating the first node to the second node; storing, by the one or more processing circuits, information based on the first timeseries, or a link to the information, in the second node responsive to an identification of the second node; identifying, by the one or more processing circuits, a third node of the plurality of nodes of the building graph representing the second air quality condition based on a second edge of the plurality of edges of the building graph relating the first node to the third node; and storing, by the one or more processing circuits, information based on the second timeseries, or a link to the information, in the third node responsive to an identification of the third node.

19. A building system of a building comprising: a mobile air quality sensor comprising at least one tractive component to move the mobile air quality sensor throughout a plurality of spaces of the building, the mobile air quality sensor configured to: collect air quality measurements of the plurality of spaces while the mobile air quality sensor moves throughout the plurality of spaces of the building, the air quality measurements comprising data identifying locations and times at which the air quality measurements were collected; and provide the air quality measurements to one or more processing circuits; the one or more processing circuits configured to: receive the air quality measurements of the plurality of spaces of the building from the mobile air quality sensor; store information, or a link to the information, in a database based on the air quality measurements; detect, based on the information, a condition occurring in a space of the plurality of spaces; and

cause, responsive to detecting the condition occurring in the space, the mobile air quality sensor to move, using the at least one tractive component, to the space to perform subsequent air quality measurements.

20. The building system of claim 19, wherein the one or more processing circuits are configured to: control operation of a piece of building equipment to control an environmental condition of at least one space of the plurality of spaces based on the information of the database.